

Bayesian Surface Habitability Mapping of Titan

from the Late Heavy Bombardment to the Solar Red-Giant Phase

A report submitted for the degree of

Master of Philosophy

Planetary Science and Life in the Universe

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All code, processed data products, and static habitability maps are openly archived at

<https://github.com/chrimblechrumble/pslu-p3>.

1 Introduction

1.1 The Search for Habitable Environments Beyond Earth

The question of whether life exists elsewhere in the universe has undergone a profound transformation over the past three decades, shifting from the domain of philosophical speculation to empirical scientific investigation. This change has been driven by two largely independent but mutually reinforcing developments: the systematic discovery that life on Earth thrives in conditions previously considered inimical to biology, and the growing recognition through robotic spacecraft exploration that chemically and thermodynamically suitable environments for life are far more widely distributed throughout the solar system than twentieth-century intuitions suggested.

The progressive characterisation of terrestrial extremophile communities has fundamentally expanded the envelope of conditions considered compatible with life. Microorganisms have been cultured from deep-sea hydrothermal vents at temperatures exceeding 121°C, from hypersaline brines with water activities as low as $a_w = 0.60$, from the anoxic pore spaces of deep crustal rocks where electrons liberated by water–rock reactions provide metabolic energy in the total absence of sunlight, and from sub-glacial Antarctic lakes that have been isolated from the atmosphere for millions of years (Cavicchioli, 2002; Rothschild and Mancinelli, 2001). Each such discovery has moved the habitable niche boundaries outwards, making it progressively more difficult to argue that any particular planetary environment is categorically uninhabitable without specific evidence that the relevant physical and chemical conditions lie entirely outside the known envelope of life.

In parallel, the outer solar system has revealed a far richer array of liquid envi-

ronments than the classical habitable zone framework would predict. The Europa detection from the Galileo mission (Pappalardo et al., 1999) and the subsequent discovery of active cryovolcanism and ocean chemistry at Enceladus (Waite et al., 2017) established that tidal heating can maintain liquid water oceans beneath ice shells at heliocentric distances far beyond any reasonable definition of the water-based habitable zone. These discoveries shifted the conceptual framework of solar system habitability from a two-dimensional picture (terrestrial planets in a narrow orbital band) to a three-dimensional one encompassing subsurface environments throughout the outer solar system.

Within this broader astrobiological context, Saturn’s moon Titan occupies a distinctive and arguably singular position. It is the only moon in the solar system possessing a substantial atmosphere — one that is denser at the surface than Earth’s and dominated by molecular nitrogen, with a structurally complex photochemistry driven by the ultraviolet flux of the distant Sun. It is the only known extraterrestrial body to host stable surface liquids (Stofan et al., 2007), completing a full precipitation-evaporation-lake cycle that is the structural analogue of the terrestrial hydrological cycle. It may have accumulated the most complex organic chemical inventory in the solar system outside Earth itself (Lorenz et al., 2008). And at depth, it contains a water–ammonia ocean whose thermodynamic conditions are increasingly well-constrained by tidal measurements and geophysical modelling. The combination of these properties makes Titan not merely a potentially habitable world but a natural laboratory for prebiotic chemistry operating at planetary scale and across a geological timescale of ~ 4.5 Gyr.

1.2 The Physical Environment of Titan

1.2.1 Atmospheric Structure and Composition

Titan’s atmosphere was first confirmed as substantially nitrogen-rich by radio occultation measurements from the Voyager 1 flyby in November 1980, which constrained the

surface pressure to approximately 1.5 bar and the surface temperature to near 90 K (Tyler et al., 1981). The subsequent *Cassini–Huygens* mission provided a comprehensive characterisation that transformed our understanding of atmospheric structure. The Huygens Atmospheric Structure Instrument (HASI) measured a surface pressure of 1.467 ± 0.001 bar and a surface temperature of 93.65 ± 0.25 K at the probe’s landing site (192.3°W , 10.6°S), with a tropospheric lapse rate of ~ 1.38 K km $^{-1}$ and a tropopause cold trap at ~ 40 km altitude where the temperature reaches a minimum of ~ 70 K (Fulchignoni et al., 2005). A stratospheric temperature inversion produces a warmer mesosphere above, analogous to the terrestrial stratosphere maintained by ozone absorption.

The *Huygens* Gas Chromatograph Mass Spectrometer (GCMS) measured a surface composition of $\approx 94.2\%$ molecular nitrogen (N_2), $5.65 \pm 0.18\%$ methane (CH_4), trace argon-40 (≈ 43 ppm, consistent with outgassing of ^{40}K from the interior), and a variety of simple organic molecules including cyanogen, acetonitrile, and propionitrile (Niemann et al., 2005). At higher altitudes, the Cassini INMS and CAPS instruments revealed a far richer chemistry: complex nitriles, benzene, and high-molecular-weight organic ions characteristic of nascent aerosol formation are detected above ~ 900 km, confirming that the tholins production process begins in the thermosphere and mesosphere and proceeds downward through the stratosphere to the surface (Waite et al., 2007).

The methane cycle in Titan’s troposphere constitutes the dominant agent of surface change and chemical transport. Methane evaporates from the north polar hydrocarbon seas, is transported equatorward by the global Hadley circulation, condenses and precipitates as rainfall over equatorial and mid-latitude terrain (Turtle et al., 2011a), and drains back towards polar basins through an extensive network of fluvial channels that Miller et al. (2021) have mapped globally from Cassini SAR data. The cycle operates on timescales ranging from individual storm events of hours to decades, through the 29.5-year Saturnian seasonal cycle, to longer-period lake-level variations driven by orbital forcing. Its significance for habitability lies in the continuous deliv-

ery of organic material from photochemically active equatorial regions to the polar lake margins, and in the chemical gradients and concentration cycles it produces at evaporating shorelines.

Ultraviolet photolysis of the N_2/CH_4 mixture in the upper atmosphere produces an extraordinary variety of complex organic photochemical products collectively termed tholins (Sagan and Khare, 1979). Laboratory synthesis of tholins from N_2/CH_4 mixtures under UV irradiation produces hundreds of identifiable molecular species including simple nitriles, amino acid precursors, nucleobase analogues, and sugar-like polyols (Cable et al., 2012). The production rate of organic aerosols is estimated at $\sim 5 \times 10^{-14} \text{ g cm}^{-2} \text{ s}^{-1}$ (Lavvas et al., 2011), implying that the surface has accumulated an organic layer estimated at between 1 and 10 km depth over solar system history (Lorenz et al., 2008). This organic reservoir is the primary chemical substrate for all habitability scenarios considered in this thesis, whether in the present epoch (as feedstock for acetylenotrophy), in the geologically recent past (as reactant in impact-melt amino acid synthesis), or in the far future (as dissolved substrate in the red-giant water–ammonia ocean).

1.2.2 Surface Geology and Geomorphology

The Cassini RADAR instrument mapped approximately 67% of Titan’s surface in synthetic aperture radar (SAR) mode at $\sim 350 \text{ m}$ per pixel resolution over 127 targeted flybys spanning 2004–2017. This coverage, whilst substantial, leaves approximately one-third of the surface — concentrated in mid-southern latitudes and portions of the northern hemisphere — dependent on lower-sensitivity ISS and VIMS imagery alone. The SAR dataset revealed a geologically diverse world operating under conditions profoundly different from any other solar system body: a geology dominated not by silicate tectonism or basaltic volcanism but by organic sediment transport, cryovolcanism, and the weathering and dissolution of ice by liquid hydrocarbon.

Lopes et al. (2019) synthesised the full Cassini SAR dataset into the first comprehensive global geomorphological map of Titan, identifying seven terrain classes that

account for all mapped surface morphologies: plains ($\sim 65\%$ of the SAR-imaged surface), dune fields ($\sim 17\%$), mountains and hummocky terrain ($\sim 8\%$), basins ($\sim 3\%$), labyrinth terrain ($\sim 4\%$), impact craters ($< 1\%$), and polar hydrocarbon lakes and seas ($\sim 1.5\%$) (Malaska et al., 2025). This geomorphological map forms a cornerstone of the habitability framework developed in this thesis, both as a source of terrain classification information for the organic abundance feature and as context for interpreting the spatial patterns in the posterior habitability maps.

The equatorial dune seas — Shangri-La, Belet, Aztlan, and Fensal, together forming a near-continuous band from roughly -30 to $+30$ latitude — are the single largest geomorphological unit on the planet. These linear dunes, oriented consistent with eastward net sand transport by the atmospheric circulation (Tokano, 2010), are composed of organic particulates, most likely complex hydrocarbons and nitriles derived from atmospheric tholin deposition (Lorenz et al., 2006). Near-infrared spectroscopy from VIMS confirms that dune terrain has the highest $1.59/1.27\text{ }\mu\text{m}$ band ratio of any terrain class on Titan, indicative of the highest tholin content (Le Mouélic et al., 2019). The dune belt accordingly receives the highest organic abundance score (0.82) in the geomorphological classification used here (Appendix C).

The north polar region above $\sim 60^\circ\text{N}$ hosts the three major hydrocarbon seas: Kraken Mare (area $\sim 400,000\text{ km}^2$, estimated maximum depth $\sim 300\text{ m}$), Ligeia Mare ($\sim 126,000\text{ km}^2$, depth $\sim 200\text{ m}$), and Punga Mare ($\sim 38,000\text{ km}^2$), together containing an estimated $\sim 70,000\text{ km}^3$ of liquid (Mastrogiuseppe et al., 2019). The compositions of the two largest seas differ measurably. Cassini active radar depth sounding and microwave emissivity measurements indicate that Ligeia Mare is predominantly methane-rich, whilst Kraken Mare is more ethane-rich (estimated at approximately 70% ethane, 16% propane, 7% methane, and 7% nitrogen by mole fraction near the 2004-era shoreline position), with composition varying seasonally as methane preferentially evaporates (Mastrogiuseppe et al., 2019). These chemical differences between adjacent sea bodies imply distinct chemical environments at their respective shorelines and will be reflected in the habitability mapping.

The Xanadu region ($\sim 90\text{--}150^\circ\text{W}$, equatorial) is anomalously radar-bright in SAR and spectrally dominated by crystalline water ice in VIMS measurements, in strong contrast to the organic-dominated signature of surrounding terrain (Brown et al., 2011). Its origin remains debated: proposed explanations include excavation of water-ice-rich crustal material by a large ancient impact (Radebaugh et al., 2011), a relict tectonic province, or anomalous ice deposition. Whatever its origin, Xanadu’s water-ice-rich composition results in a low organic abundance score and consequently low habitability score in the present model.

Ontario Lacus at the south pole (179°W , 72°S) is the only confirmed southern-hemisphere lake (Turtle et al., 2009), with an estimated area of $\sim 15,000\text{ km}^2$. ISS observations revealed measurable shoreline recession between 2005 and 2010, suggesting net evaporation under southern summer insolation at the Cassini epoch. Its significantly smaller size and lower associated methane cycle activity compared to the north polar seas means it scores notably lower in the present-epoch habitability map.

1.2.3 Interior Structure and the Subsurface Ocean

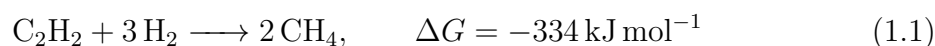
The most critical interior constraint for habitability purposes is the tidal Love number $k_2 = 0.589 \pm 0.150$ measured by Iess et al. (2012) from orbital perturbations of the Cassini spacecraft during close flybys. This value greatly exceeds the prediction for a fully solid (rocky core plus solid ice shell) body, for which $k_2 < 0.03$. The only viable physical explanation is a global subsurface liquid layer that deforms readily under Saturn’s tidal force, amplifying the tidal response. Geophysical interior models constrain this layer to a water–ammonia ocean at $\sim 55\text{--}80\text{ km}$ depth, beneath an outer ice-I shell and overlying an ice-VI mantle and silicate core (Nimmo and Bills, 2010; Tobie et al., 2005). The ocean’s ammonia concentration is estimated at 5–15 wt% from density constraints (Mitri et al., 2014); this concentration depresses the melting point of the mixture to near the water–ammonia eutectic (176 K at 32.6 wt% NH_3 ; Grasset and Pargamin 2005), ensuring the ocean remains liquid at its current depth despite surface temperatures of only 94 K.

The subsurface ocean has direct habitability relevance for two reasons. First, [Affholder et al. \(2025\)](#) demonstrated that glycine fermentation ($\Delta G \approx -20$ to -50 kJ/mol under estimated ocean conditions) is energetically viable within the ocean, and that the dissolved organic nitrogen available from hydrolysed tholins provides adequate substrate for simple metabolic cycles. Second, at the surface, SAR-bright annular features surrounding several large impact craters are interpreted as ancient melt-water fronts where liquid water sourced from the subsurface ocean (or from impact melting) spread outward from crater basins, potentially bringing ocean-derived chemistry into contact with the surface organic inventory ([Wood et al., 2010](#)). This interpretation underpins the elevated subsurface ocean proximity score at Selk crater discussed in Chapter 3.

1.3 Chemical Habitability of Titan’s Surface: Prior Theoretical Work

1.3.1 The Acetylenotrophy Hypothesis

The first quantitative assessment of the energetic viability of life on Titan’s surface was provided by [McKay and Smith \(2005\)](#), who evaluated the metabolic energy available from the combination of atmospheric acetylene (C_2H_2) with molecular hydrogen (H_2) — a process they termed acetylenotrophy:



Acetylene is produced continuously by ultraviolet photolysis of methane in the upper atmosphere and sediments to the surface. Molecular hydrogen, a photochemical by-product, similarly settles downward. McKay and Smith calculated that the available acetylenotrophy energy flux at Titan’s surface is comparable to that available from photosynthesis at the deep ocean floor on Earth, and substantially exceeds the metabolic energy requirements of the most energy-sparse known terrestrial organisms,

including methanogens and anaerobic chemoautotrophs.

Possible observational support for this hypothesis emerged from [Strobel \(2010\)](#), who reported a downward flux of H_2 to Titan's surface of $\sim 10^{10} \text{ molecules cm}^{-2} \text{ s}^{-1}$ — approximately 100 times the flux predicted by known abiotic surface reactions — based on analysis of Cassini INMS data. A surface sink for H_2 exceeding abiotic predictions is precisely consistent with acetylenotrophy. However, [Teolis et al. \(2015\)](#) subsequently identified INMS calibration uncertainties at low signal levels that could affect the reported flux magnitude, and the measurement must be considered tentative pending confirmation by future in-situ measurements. The energetics of acetylenotrophy were revisited using updated thermochemical data by [Yanez et al. \(2024\)](#), who calculated a free energy yield of $69\text{--}78 \text{ kJ mol}^{-1} \text{ C}$ under Titan surface conditions — somewhat more conservative than the original McKay & Smith estimate but still comfortably above the minimum for terrestrial methanogens ($\sim 25\text{--}65 \text{ kJ mol}^{-1} \text{ C}$).

1.3.2 Non-Aqueous Biochemistry and the Azotosome Hypothesis

A complementary line of theoretical work has examined the possibility of life operating with liquid methane as its biochemical solvent rather than water. [Benner et al. \(2004\)](#) provided a theoretical basis for non-aqueous biochemistry, arguing that whilst water's polarity and hydrogen-bonding capacity are indeed important for terrestrial biochemistry, these properties are not thermodynamically fundamental requirements; a sufficiently different but internally consistent biochemistry might evolve under different solvent conditions. In liquid methane at $\sim 90 \text{ K}$, the principal structural challenge is membrane formation: conventional phospholipid bilayers, whose amphiphilic nature depends on the polarity contrast between water and non-polar lipid tails, would not form spontaneously in a non-polar solvent.

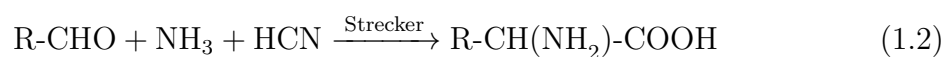
[Stevenson et al. \(2015\)](#) addressed this challenge through molecular dynamics simulations, showing that small nitrogen-containing molecules such as acrylonitrile (CH_2CHCN), subsequently detected in Titan's stratosphere by ALMA ([Palmer et al., 2017](#)), can

spontaneously form stable membrane-like vesicles (termed azotosomes) in liquid methane. These azotosome membranes would have reversed polarity relative to terrestrial cell membranes, with polar head groups directed inward towards a polar interior. Whilst no experimental demonstration of azotosome-based metabolism exists, the thermodynamic and structural feasibility of membrane formation establishes at least one plausible path to cellular compartmentalisation in a methane-solvent biology.

1.3.3 Prebiotic Chemistry in Post-Impact Melt Environments

A third and distinctly different habitability pathway on Titan involves transient liquid water rather than the permanent hydrocarbon lakes. When large impactors strike Titan’s ice-rich crust, the kinetic energy of impact melts the surrounding material, producing water–ammonia–organic mixture ponds that persist at liquid temperatures for 10^3 – 10^4 years before refreezing (O’Brien et al., 2005). For a 50 km diameter rocky impactor at Titan’s escape velocity of 2.64 km s^{-1} , the delivered kinetic energy is sufficient to melt $\sim 10^4 \text{ km}^3$ of ice. During the Late Heavy Bombardment the impactor flux was approximately 100–1000 times the present background rate, ensuring that multiple active melt pools existed simultaneously on Titan’s surface at any time during the $\sim 300 \text{ Myr}$ duration of the bombardment.

Madan et al. (2026) demonstrated through thermodynamic and kinetic modelling that the Strecker reaction:



is accessible in post-impact melt environments at temperatures of 200–350 K. All three precursors are available: surface aldehydes (R-CHO) from tholin decomposition, NH_3 from the water–ammonia melt, and HCN from the nitrogen-rich atmosphere (HCN is one of the most abundant photochemical products in Titan’s stratosphere; Vuitton et al. 2007). The amino acid products are stable on timescales comparable to the melt pool lifetime. This pathway is central to the astrobiological rationale for the

Dragonfly mission’s primary landing site at Selk crater, where the post-impact melt chemistry of an ancient large impact is preserved in the geological record.

1.3.4 The Far-Future Habitability Window

Lorenz et al. (1997) identified a fourth temporal window of classical water-based habitability on Titan associated with the Sun’s eventual departure from the main sequence. As solar luminosity increases by factors of hundreds over $\sim 10^7$ – 10^8 yr during the red-giant phase, Titan’s surface temperature will rise through the water–ammonia eutectic (176 K) and remain above it for an estimated 200–500 Myr (Lorenz et al., 1997; Schröder and Connon Smith, 2008). During this window, Titan’s entire surface would be covered by a global water–ammonia ocean, overlying an organic sediment reservoir that by that epoch will have accumulated for ~ 16 Gyr of continuous atmospheric deposition — a chemical stockpile of extraordinary richness. In this scenario, Titan would become transiently and potentially uniquely the most habitable body in the solar system, at a time when Earth and Mars will have been sterilised by the expanding solar photosphere. The quantitative characterisation of this future epoch is one of the novel contributions of this thesis.

1.4 The Cassini–Huygens Mission and Its Astrobiological Legacy

The joint NASA/ESA *Cassini–Huygens* mission operated in the Saturnian system from July 2004 until September 2017, when the spacecraft was deliberately deorbited into Saturn’s atmosphere to prevent biological contamination of potentially habitable moons. The 13-year operational lifetime enabled 127 targeted Titan flybys, the *Huygens* probe descent and landing in January 2005, and a systematic multi-instrument survey of Titan’s atmosphere, surface, and interior that constitutes the observational foundation of this thesis.

The observational coverage is comprehensive but incomplete. SAR mapping reached

~67% of the surface; the VIMS near-infrared spectral coverage relevant to organic abundance assessment reaches approximately 50%; the CIRS thermal mapping achieved global coverage at ~10° latitude resolution; and the radar altimetry dataset covers ~90% of the globe after global interpolation. The GCMS aboard *Huygens* provided the single direct measurement of surface chemical composition, at one equatorial site. This pattern — broad but spatially uneven remote sensing coverage combined with a single point of in-situ ground truth — defines both the strengths and the principal limitations of the quantitative habitability framework developed here.

The post-*Cassini* synthesis volumes *Titan After Cassini-Huygens* (Cambridge University Press, 2025) have distilled the scientific legacy of the mission. In the context of surface habitability, the synthesis highlights three significant gaps: (1) the absence of any quantitative spatially-resolved habitability probability map of the full globe; (2) the absence of an explicit multi-epoch temporal treatment connecting the LHB, present, and far-future habitability regimes; and (3) the absence of a reproducible, openly archived analysis framework integrating the full suite of Cassini data modalities. This thesis addresses all three.

1.5 The Dragonfly Mission

NASA’s *Dragonfly* mission (Lorenz et al., 2018) was selected as the fourth New Frontiers programme mission in June 2019. It is currently en route to Titan with a projected arrival in 2034. The mission deploys a dual-quadcopter rotorcraft that exploits Titan’s dense atmosphere (surface density $\rho \approx 5.4 \text{ kg m}^{-3}$) and low gravity to achieve powered flight, enabling mobility impossible for wheeled rovers. The baseline mission plans for ~175 km of total traverse distance over a 2.7-year nominal mission, beginning in the Belet dune field to the north-west of Selk crater and progressing through the crater ejecta blanket to the crater floor.

The scientific rationale for Selk as the primary target rests on the exposure of water-ice-rich crustal material in the ejecta (enabling Strecker amino acid synthesis when contacted by melt water), the adjacent organic-rich dune terrain, and the radar-

bright crater rim morphology consistent with ancient liquid-water exposure. *Dragonfly* carries a mass spectrometer (DraMS), a gamma-ray and neutron spectrometer (DraGNS), a drill, a camera suite (DragonCam), and a meteorology package. The analysis in this thesis provides a quantitative evaluation of the mission’s site selection and identifies which habitability features each instrument is best positioned to test.

1.6 Scope and Research Questions

Several preceding studies have attempted to place planetary habitability on a quantitative footing. The *Earth Similarity Index* (Schulze-Makuch et al., 2011) scores planetary bodies against Earth on temperature, density, escape velocity, and surface chemistry, but produces a single scalar per body and cannot represent spatial or temporal variation. Hendrix et al. (2019) reviewed the habitability potential of icy ocean worlds using qualitative multi-criteria matrices, providing rich scientific context but no spatial resolution or probabilistic framework. Logistic regression and machine-learning habitability classifiers (e.g. Spiegel and Turner 2012) have been applied to exoplanet populations but require labelled training data that does not exist for Titan. The Bayesian approach adopted here differs from all of these: it encodes scientific prior knowledge as probability distributions, updates them with observational data via conjugate updating, and returns credible intervals rather than point estimates — making uncertainty explicit rather than implicit. The choice of a Beta-conjugate (Beta-Binomial) model is motivated by the bounded $[0, 1]$ domain of habitability probability and by the analytical tractability that permits simultaneous evaluation at all 6.49 million grid pixels without numerical integration or MCMC. A Gaussian process or multivariate logistic regression could in principle capture feature correlations more richly, but would require either strong covariance priors or prohibitively dense labelled sampling to estimate the correlation structure from Cassini data alone. The Beta-conjugate approach is preferred as the most tractable framework that preserves physical interpretability at every pixel.

Previous habitability assessments of Titan have shared four important limitations.

They are predominantly qualitative or point-specific, lacking global spatial resolution. They treat the present epoch as the only relevant state, ignoring both the LHB prebiotic-chemistry window and the far-future red-giant window. They do not integrate the full suite of Cassini data modalities into a unified probabilistic framework. And they lack the reproducibility afforded by an openly archived analysis pipeline.

The closest methodological precedent is [Affholder et al. \(2021\)](#), who applied Bayesian statistical inference to Cassini INMS plume data from Enceladus, comparing the observed methane and hydrogen escape rates against models of abiotic serpentinisation and biotic methanogenesis. Their finding — that the observed methane escape rate is most parsimoniously explained by biological methanogenesis if the prior probability of life is sufficient — represents the most quantitatively rigorous Bayesian biosignature assessment yet published for a solar system body. The present work is directly inspired by this approach, extended from a single-point plume observation to a full-globe, multi-feature, multi-epoch spatial mapping.

The specific research questions addressed are:

- RQ1** *What is the spatial distribution of surface habitability on present-day Titan, and which terrain types, geological features, and atmospheric zones produce the highest scores? Are the data-derived feature importances consistent with the physically motivated prior weights?*
- RQ2** *How did habitability differ during the Late Heavy Bombardment, when impact-melt ponds provided transient liquid water? Which impact structures were the most astrobiologically significant, and what are the implications for prebiotic chemistry?*
- RQ3** *How does habitability evolve through the temporal sequence from the LHB to the red-giant future? Is Titan potentially the most habitable body in the solar system during the red-giant ocean window?*
- RQ4** *Does the derived habitability map independently validate the Dragonfly landing site at Selk crater, and which instruments are best positioned to test the model's*

predictions?

Chapter 2 presents the data sources, feature extraction, Bayesian inference model, and temporal methodology. Chapter 3 presents and interprets the habitability maps at each epoch, the feature importances, the temporal evolution, and the detailed Selk crater analysis. Chapter 4 discusses the *Dragonfly* mission implications, biosignature framework, limitations, and future work.

2 Methods

2.1 Overview of the Analytical Approach

2.1.1 Acronyms and Nomenclature

The following acronyms are used throughout this chapter and the remainder of the thesis. They are defined here for convenience and also at the point of first use.

SAR	Synthetic Aperture Radar. An active microwave imaging technique in which the motion of the spacecraft is used to synthesise a very long antenna aperture, yielding fine-resolution imagery of surface texture and composition.
VIMS	Visible and Infrared Mapping Spectrometer. The <i>Cassini</i> instrument that imaged Titan’s surface through six near-infrared atmospheric transparency windows, providing surface composition data.
CIRS	Composite Infrared Spectrometer. The <i>Cassini</i> thermal emission instrument used here to constrain the surface temperature field.
GTDR	Global Titan Digital elevation model, Radar — the raw Cassini RADAR altimetry dataset archived in the Planetary Data System.
GTDE	Global Titan Digital Elevation model — the globally interpolated topographic product of Corlies et al. (2017) derived from the GTDR and supplementary data sources.
DEM	Digital Elevation Model. A gridded representation of surface topography.

PDS	Planetary Data System. The NASA archive for planetary mission data.
HDI	Highest Density Interval. A Bayesian credible interval containing the specified probability mass within the narrowest possible range.

2.1.2 Notation: Feature Weights and Prior Means

Two quantities appear throughout this chapter and are defined here to avoid repetition. Each habitability-proxy feature f_i ($i = 1, \dots, 8$) is assigned:

- A **weight** $w_i \in [0, 1]$, where $\sum_i w_i = 1$. The weight encodes the relative astrobiological importance of feature i as a habitability indicator. A feature with a higher weight exerts a larger influence on the posterior probability of habitability, all else being equal. Weights are assigned from physical reasoning and validated against data-derived importances (Section 3.2.5).
- A **prior mean** $\mu_i \in [0, 1]$. This is the expected spatially-averaged value of feature i before any observational data are considered — a statement of our prior knowledge about how common or widespread the habitability-relevant condition is on Titan. For example, liquid hydrocarbon covers $\sim 2\%$ of Titan’s surface (Hayes et al., 2008), so its prior mean is $\mu_1 = 0.02$. The weighted sum of all prior means defines the global prior mean habitability: $\mu_0 = \sum_i w_i \mu_i = 0.331$. Note that the value $\mu_0 = 0.331$ does not appear in the prior literature as a global habitability estimate for Titan; it emerges from the component prior means in Table B.1 and should be interpreted as encoding the observation that Titan’s surface is globally organic-rich whilst simultaneously recognising that liquid solvent is present at only $\sim 2\%$ of the surface. The resulting moderate prior is intentionally revisionable: the weak concentration parameter $\kappa = 5$ ensures that the data, not the prior, determine the posterior at any pixel where observational coverage is good. Full values are tabulated in Appendix B.

This notation first appears in a practical context in Section 2.3, where the eight features are described individually. Their role in the Bayesian update is explained in Section 2.4.

2.1.3 The Bayesian Approach: Motivation and Structure

The central methodological challenge in quantifying Titan’s surface habitability is that no single Cassini instrument directly measures habitability. Each instrument provides a partial and indirect proxy: radar backscatter reveals surface composition and liquid surfaces; near-infrared spectroscopy constrains organic content; thermal emission maps the temperature field that drives the methane cycle; altimetry constrains topography and hence organic accumulation patterns; and the geomorphological classification synthesises the collective information from all instruments into terrain types whose habitability implications can be grounded in laboratory and theoretical work.

The approach taken here is to combine these partial proxies through a Bayesian inference framework that produces, at each grid cell, a posterior probability of habitability $P(H \mid \mathbf{f})$ given the vector of observed features $\mathbf{f} = (f_1, \dots, f_8)$. This is structurally analogous to the approach of [Affholder et al. \(2021\)](#), who used Bayesian methods to evaluate whether the observed methane flux in Enceladus’s plumes is more consistent with abiotic serpentinisation or with biological methanogenesis. The key extensions here are spatial resolution (a 6.49-million-pixel global map rather than a single-point estimate), the number of independent observational inputs (eight features rather than one), and the temporal scope (five epochs spanning geological time rather than a single snapshot).

The structure of the inference is: (1) encode prior knowledge about Titan’s habitability into a probability distribution over the latent variable H ; (2) update that distribution using each observed feature f_i as evidence; and (3) report the resulting posterior probability $P(H \mid \mathbf{f})$ as the habitability estimate at each pixel. The mathematical details of this procedure are given in Section 2.4; the present section focuses on the observational inputs.

The analysis is applied at 4490 m/px resolution on a canonical equirectangular grid covering the full globe (1802 pixels in latitude $\times$ 3603 pixels in longitude), using a west-positive longitude convention consistent with all Cassini raster products. This resolution is chosen to match the coarsest primary dataset (the GTDR altimetry mosaic at ~ 22 km per pixel) whilst resolving lake margins and fluvial features at ~ 45 km scale. All eight input datasets are reprojected to this common grid before feature extraction.

Figure 2.1 provides a schematic overview of the five-stage analytical pipeline: data acquisition, preprocessing to the canonical grid, feature extraction, Bayesian inference, and visualisation.

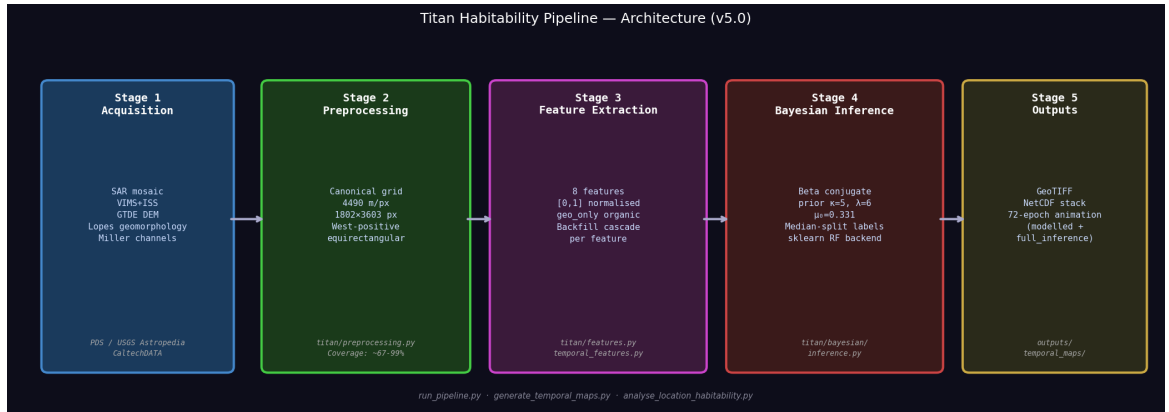


Figure 2.1: Schematic overview of the five-stage Titan habitability analysis pipeline. All eight habitability-proxy features are extracted in Stage 3 and combined in the Bayesian posterior update of Stage 4 to produce $P(H | \mathbf{f})$ maps at each of the five temporal configurations.

2.2 Observational Data Sources

2.2.1 Cassini RADAR Synthetic Aperture Radar Mosaic

The Cassini RADAR instrument operated in SAR mode during Titan flybys, imaging the surface at ~ 350 m per pixel over the 127 targeted flybys from 2004 to 2017 (Janssen et al., 2016). SAR measures the normalised radar cross-section σ^0 (typically expressed in dB), which is sensitive to surface roughness, dielectric properties, and the presence of smooth or conducting materials. Low backscatter ($\sigma^0 < -20$ dB) is

diagnostic of smooth surfaces (consistent with liquid hydrocarbon lakes) or dielectrically absorbing material (consistent with tholin-rich dune sand). High backscatter correlates with rough, water-ice-rich material such as mountain terrain, crater ejecta, and some cryovolcanic surfaces. The SAR mosaic covers $\sim 67\%$ of Titan’s surface and constitutes the primary data source for the liquid hydrocarbon, chemical energy, and subsurface ocean features.

2.2.2 VIMS Near-Infrared Spectral Mosaic

The Cassini Visible and Infrared Mapping Spectrometer (VIMS) observed Titan’s surface through atmospheric transparency windows at 0.94, 1.08, 1.27, 1.59, 2.03, and 2.79 μm . The band ratio 1.59/1.27 μm is an established proxy for tholin (complex organic) abundance at the surface: the 1.59 μm window probes deeper into the tholin-bearing surface whilst the 1.27 μm window provides atmospheric normalisation (Barnes et al., 2007). The Seignovert et al. (2019) combined VIMS+ISS mosaic (Seignovert et al., 2019) provides this band ratio map at 2–4 km per pixel coverage, but only for the 0–180°W hemisphere ($\sim 50\%$ of the globe). The significance of this coverage gap for the organic abundance feature is discussed in Section 2.3.2.

2.2.3 CIRS Thermal Emission Observations

The Composite Infrared Spectrometer (CIRS) measured Titan’s surface brightness temperature at far-infrared wavelengths throughout the Cassini mission. Jennings et al. (2019) derived an analytical model of the latitude-dependent surface temperature from 13 years of CIRS observations, producing the formula:

$$T(\phi, Y) = (93.53 - 0.095 Y) \cos\left[(\phi + 0.85 - 3.2 Y)(0.0029 - 0.00006 Y)\right] \quad (2.1)$$

where ϕ is latitude in degrees and $Y = t_{\text{CE}} - 2009.61$ is years from the northern spring equinox (22 May 2009). This formula reproduces the observed zonal surface temperatures to within ± 0.4 K (rms) across the full Cassini epoch and provides 100% global

coverage, grounded in CIRS focal-plane-1 limb and nadir observations across the full 2004–2017 mission. The model captures only latitudinal and seasonal variation; no longitudinal inhomogeneity is represented (consistent with the absence of significant longitudinal temperature gradients in the Cassini dataset). For temporal epochs other than the present, the mean annual temperature field is held stationary: this approximation is justified because the dominant habitability driver at non-present epochs is the liquid-hydrocarbon scale function $s_1(t)$ rather than the temperature gradient, and the methane-cycle feature contributes only $w_4 = 0.15$ to the posterior. It is used to synthesise the global temperature field and its meridional gradient, which enters the methane cycle feature (Section 2.3.4).

2.2.4 GTDE/GTDR Topographic Models

The Global Titan Digital Elevation model (GTDE), produced by [Corlies et al. \(2017\)](#), integrates Cassini RADAR altimetry, SARtopo stereophotogrammetry, and radial basis function interpolation to produce a globally-interpolated topographic model at ~ 22 km per pixel (2 pixels per degree). Coverage is approximately 90% of the globe, with the south polar region and some high-latitude northern areas relying most heavily on interpolation. The topographic model contributes to the chemical energy feature (identifying topographic lows as preferential organic accumulation sites), the topographic complexity feature (through local roughness estimation), and the surface–atmosphere interaction feature (through slope calculation).

2.2.5 Lopes (2019) Global Geomorphological Map

The global geomorphological map of [Lopes et al. \(2019\)](#) classifies the Cassini SAR-imaged surface into seven terrain units (plains, dunes, mountains, basins, labyrinth terrain, craters, and lakes) using SAR texture, backscatter, morphology, and VIMS spectral correlation. This map provides the terrain classification that enters the organic abundance feature (as a globally seamless gap-fill) and the geomorphological diversity feature. The shapefiles are provided in east-positive geographic coordinates

and require conversion to the west-positive convention of the analysis grid.

2.2.6 Miller (2021) Global Fluvial Channel Map

Miller et al. (2021) identified and mapped Titan’s global fluvial channel network from Cassini SAR data, producing a vector dataset of channel locations and widths. This network constitutes the primary mechanism for organic material transport from equatorial source regions to the polar basins, and channel density is used as one component of the surface–atmosphere interaction feature.

2.2.7 Tidal Love Number and Gravity Data

The degree-2 tidal Love number $k_2 = 0.589 \pm 0.150$ from Iess et al. (2012) provides a global constraint on interior structure, confirming the presence of a subsurface liquid layer. This scalar quantity enters as a global floor value in the subsurface ocean proximity feature.

2.2.8 Hedgepeth (2020) Global Crater Catalogue

The global catalogue of confirmed and probable impact structures on Titan by Hedgepeth et al. (2020) records crater diameters and positions for all resolved impact features in the Cassini SAR dataset. This catalogue is used exclusively in the PAST epoch to generate the impact-melt proximity feature.

2.3 Feature Extraction and Scientific Basis

Eight habitability-proxy features are extracted from the above data sources. Figure 2.2 shows all eight feature maps at the present epoch, illustrating their distinct spatial patterns. Figure 2.3 shows how the relative activity of each feature is modulated across geological time.

All features are normalised to the range $[0, 1]$ using a robust percentile-clipping

approach:

$$\hat{f}_i = \text{clip}\left(\frac{f_i - f_{i,p2}}{f_{i,p98} - f_{i,p2}}, 0, 1\right) \quad (2.2)$$

where $f_{i,p2}$ and $f_{i,p98}$ are the 2nd and 98th percentiles of finite values in the global layer, mapping the central 96% of the observed dynamic range onto $[0, 1]$ whilst preventing extreme outliers from compressing the scale. Where a feature is unavailable at a particular pixel (due to data gaps), the value is set to NaN and handled through the Bayesian missing-data imputation procedure described in Section 2.4.5.

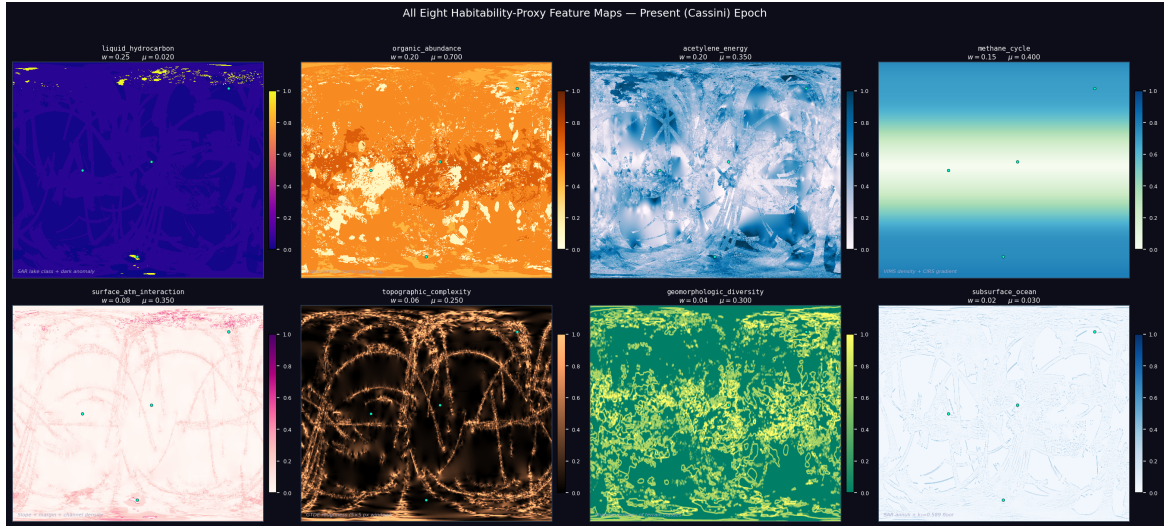


Figure 2.2: All eight habitability-proxy feature maps at the present (Cassini) epoch, each normalised to $[0, 1]$. The panels illustrate the distinct spatial patterns contributed by each feature: the polar concentration of liquid hydrocarbon (f_1), the near-global organic abundance (f_2), the topographically-modulated chemical energy proxy (f_3), and the latitude-dependent methane cycle signal (f_4), amongst others. The combined posterior is derived from all eight simultaneously via the Bayesian update described in Section 2.4. Prior weight w_i and prior mean μ_i are shown for each panel.

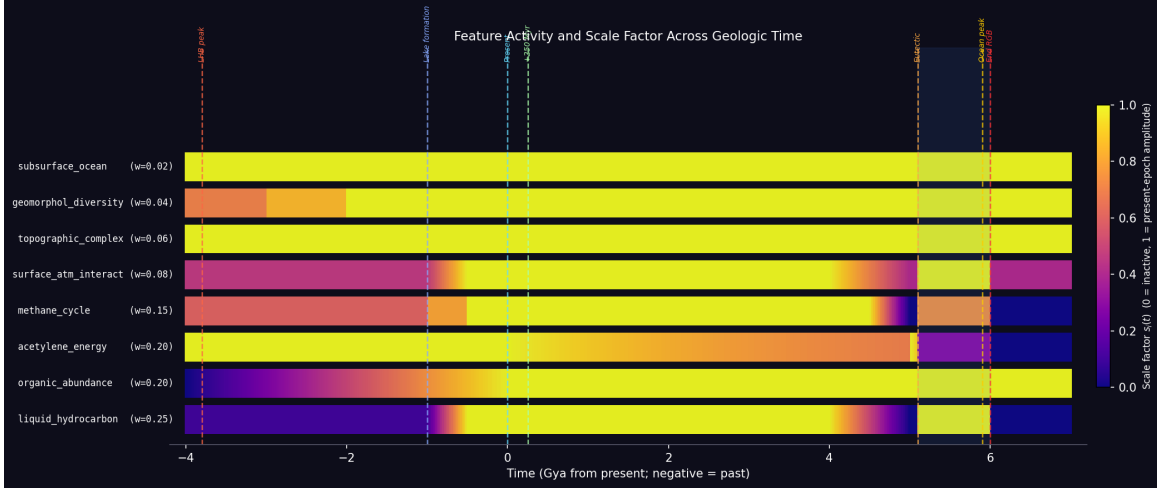


Figure 2.3: Scale factor (relative activity) for each feature as a function of geological time from -4.0 Gya to $+7.0$ Gya. Colour intensity encodes the scale applied to each feature’s prior mean at that epoch: bright yellow indicates full present-epoch amplitude; dark blue indicates near-zero activity. Key events (LHB peak, lake-formation onset, present epoch, solar warming ramp, eutectic crossing, ocean peak) are marked with vertical dashed lines. The impact-melt proxy and cryovolcanic flux features are active only in pre-lake epochs and are shown in the lower two rows.

2.3.1 Feature 1: Liquid Hydrocarbon Availability ($w_1 = 0.25$, $\mu_1 = 0.02$)

Scientific basis. Liquid methane and ethane are the proposed non-aqueous biochemical solvents for any life operating on Titan’s present-day surface (McKay, 2016; McKay and Smith, 2005). At the ambient surface temperature of ~ 94 K, water is structurally as rigid as rock and cannot support molecular diffusion processes fundamental to metabolism. Liquid methane and ethane are, however, stable at 94 K and 1.47 bar, as confirmed by the permanent north polar seas. Benner et al. (2004) proposed that non-polar solvents can in principle support biochemistry through different but self-consistent molecular architectures, and Stevenson et al. (2015) demonstrated through molecular dynamics simulations that acrylonitrile molecules present in Titan’s atmosphere can form stable vesicle membranes in liquid methane. The presence of persistent liquid hydrocarbon is therefore treated as the single most important habitability prerequisite for surface life at the present epoch, and accordingly receives the highest prior weight ($w_1 = 0.25$).

Prior mean. The north polar seas cover approximately 1.5–2% of Titan’s surface (Hayes et al., 2008), giving a prior mean of $\mu_1 = 0.02$. Despite the high weight, the low prior mean means that most pixels contribute negligibly through this feature; its habitability impact is strongly localised to the polar regions.

Implementation. The primary source is the Lopes et al. (2019) lake polygon class (terrain class 7), rasterised to the canonical grid as a binary lake mask. A secondary contribution comes from SAR backscatter dark anomalies not otherwise attributed to organic material. Confirmed empty lake basins from the Birch et al. (2017) analysis are explicitly assigned $f_1 = 0$ to reflect the absence of present-day liquid even within geomorphologically defined basins. In the absence of SAR coverage, ISS 938 nm albedo imagery provides a proxy, with smooth, low-albedo regions consistent with liquid surfaces elevated towards higher feature values.

Backfill. In regions lacking both SAR and ISS coverage, the feature is assigned NaN and imputed at the prior mean during inference.

2.3.2 Feature 2: Organic Abundance ($w_2 = 0.20$, $\mu_2 = 0.70$)

Scientific basis. Tholins — the complex refractory macromolecular organic compounds produced continuously by ultraviolet photolysis of the N_2/CH_4 mixture in the upper atmosphere — are the dominant surface material across most of Titan. Their astrobiological significance is multifaceted. Laboratory hydrolysis of tholins in liquid water at biologically relevant temperatures (273–373 K) produces amino acids, nucleobase analogues, and sugar-like polyols at yields of $\sim 0.1\%$ by mass (Neish et al., 2018), directly confirming that tholins can serve as a chemical feedstock for prebiotic synthesis when liquid water is available — either from impact melts at past epochs or from the red-giant ocean in the far future. Meyer-Dombard et al. (2025) characterise the tholin layer as the primary chemical substrate for any surface habitability scenario on Titan. The feature weight ($w_2 = 0.20$) is second only to liquid hydrocarbon, reflecting this central role.

Prior mean. The revised prior mean of $\mu_2 = 0.70$ (updated from 0.60 in earlier pipeline versions) reflects updated spectral characterisation from [Malaska et al. \(2025\)](#) and [Cable et al. \(2012\)](#), indicating that approximately 82% of the SAR-classified surface by area falls into organic-rich terrain classes (dune: 17%, plains: 65%). The organic abundance feature consequently makes the largest single contribution to the global prior mean ($w_2\mu_2 = 0.140$).

Implementation and the seamless organic mode. The primary spectral data source is the VIMS+ISS 1.59/1.27 μm band-ratio mosaic from [Seignovert et al. \(2019\)](#), but this covers only the 0–180°W hemisphere ($\sim 50\%$ of the globe). Extensive testing of direct blending between the VIMS band ratios and geomorphological class scores in the 180–360°W hemisphere produced a visible seam artefact at the 180°W boundary: the absolute calibration of the VIMS spectral ratio differs systematically from the terrain-class score at the coverage boundary, producing a step change of ~ 0.13 in median organic abundance clearly visible in rendered maps. This artefact cannot be eliminated by a single global offset because the boundary region terrain is not representative of all terrain classes. The analysis therefore uses a seamless geomorphological mode, applying the terrain-class scores from Table C.1 (Appendix C) globally. Whilst this approach sacrifices per-pixel spectral variation within the VIMS-covered hemisphere (reducing the standard deviation from 0.27 for VIMS data to 0.16 for terrain-class scores), it ensures that no spatial artefact from the uneven VIMS coverage contaminates the habitability maps.

Backfill. The Lopes (2019) terrain classification covers $\sim 90\%$ of the SAR-imaged surface ($\sim 67\%$ of the globe). In regions outside SAR coverage, terrain class is estimated from ISS albedo units with reduced specificity.

2.3.3 Feature 3: Chemical Energy Availability ($w_3 = 0.20$, $\mu_3 = 0.35$)

Scientific basis. Life requires a sustained source of chemical free energy to drive metabolic reactions against thermodynamic equilibrium. On Titan’s surface, the primary candidate is acetylenotrophy (Eq. 1.1), with $\Delta G = -334 \text{ kJ mol}^{-1}$ (where ΔG is the Gibbs Free Energy change of the reaction, a negative value indicating spontaneous energy release) available from acetylene–hydrogen combination (McKay and Smith, 2005). Yanez et al. (2024) recalculated the available energy under updated thermochemical conditions at Titan’s surface as 69–78 kJ mol^{−1} C, well above the minimum for terrestrial methanogens. Acetylene is produced continuously by UV photolysis of methane and sediments to the surface where it has been directly detected by GCMS (Niemann et al., 2005). The chemical energy feature is therefore a proxy for the co-availability of acetylene and hydrogen at the surface.

Implementation. Since the acetylene surface abundance cannot be directly mapped by any Cassini instrument, it is approximated through two components. SAR backscatter (weight 70% where coverage exists): SAR-dark surfaces characteristic of organic-rich terrain represent regions where acetylene is most likely to accumulate without competing abiotic reactions consuming it rapidly. Topographic inversion (weight 30%): acetylene and other atmospheric condensates preferentially accumulate in topographic lows through both gravitational settling and fluvial transport; the normalised negative of the GTDE elevation ($-z$) assigns high values to topographic lows. In SAR-absent regions, the feature reverts to the pure topographic inversion component.

2.3.4 Feature 4: Methane Cycle Intensity ($w_4 = 0.15$, $\mu_4 = 0.40$)

Scientific basis. The methane cycle is the dominant agent of chemical transport on Titan. It delivers organic material from equatorial photochemical production regions

to the polar lake margins, maintains chemical gradients at shorelines, and provides seasonal wet–dry cycling at lake edges that some prebiotic chemistry scenarios identify as important for concentrating organic reactants and driving condensation reactions (Benner et al., 2004; Mitchell and Lora, 2016). Regions experiencing the most active methane cycling receive fresh organic inputs most regularly and maintain the most dynamic chemical environments.

Implementation. The methane cycle intensity is a weighted composite of three independently computed components:

1. **VIMS observation density** (50%): the number of VIMS spectral footprints per grid cell, smoothed with a 5-pixel Gaussian kernel. Cassini’s VIMS observation targeting was scientifically motivated, prioritising regions of active methane-cycle interest; higher coverage density therefore correlates with active methane cycling zones (Le Mouélic et al., 2019).
2. **Meridional temperature gradient** $|\partial T/\partial \phi|$ (25%): derived from the Jennings (2019) model (Eq. 2.1), evaluated at $Y = +1.39$ (year 2011.0, the approximate mission midpoint). Regions where the surface temperature gradient is steepest experience the strongest differential evaporation and precipitation, driving the most vigorous methane transport.
3. **Latitude prior** (25%): the function $\exp[-(\phi \pm 45)^2/(25)^2]$, following Mitchell and Lora (2016)’s general circulation modelling result that the mid-latitude regions host the primary methane cycle activity during the Cassini epoch.

This feature has 100% global coverage by construction and requires no backfill.

2.3.5 Feature 5: Surface–Atmosphere Interaction Intensity

$$(w_5 = 0.08, \mu_5 = 0.35)$$

Scientific basis. The physical interface between surface and atmosphere is where gas-phase photochemical products encounter potential liquid solvents and organic

substrates. Lake margins and fluvial channel termini are the primary loci of this exchange. [Hayes \(2016\)](#) describes the complex sedimentological and chemical processes at hydrocarbon lake shorelines, including the evaporation-driven concentration of dissolved organics and the seasonal flooding and drying cycles that expose organic sediments to liquid contact. [Miller et al. \(2021\)](#) documents that the fluvial channel network terminates preferentially at polar lake margins, establishing these as zones of maximum organic delivery and chemical gradient activity.

Implementation. A weighted sum of three spatial components:

$$f_5 = 0.30 \hat{s} + 0.40 \hat{m} + 0.30 \hat{c} \quad (2.3)$$

where $\hat{s}$ is the normalised topographic slope $|\nabla z|$ from the GTDE DEM (high slopes drive physical transport and mixing); $\hat{m}$ is a binary lake-margin mask extended to ~ 45 km from the nearest confirmed lake shore; and $\hat{c}$ is the Gaussian-smoothed ($\sigma = 3$ pixels) channel density from the [Miller et al. \(2021\)](#) dataset, normalised at the 2nd–98th percentile range. The lake-margin component receives the highest within-feature weight (40%) because shorelines concentrate all three relevant processes — liquid contact, organic delivery, and thermal/chemical cycling.

2.3.6 Feature 6: Topographic Complexity ($w_6 = 0.06$, $\mu_6 = 0.25$)

Scientific basis. Local topographic roughness at ~ 20 km scale correlates with the dissected terrain types that most frequently expose water-ice-bearing crustal material ([Lopes et al., 2019](#)). On Earth, topographic heterogeneity at ecologically relevant scales creates micro-environmental diversity — variations in drainage, aspect, moisture, and chemical exposure — that underpins elevated species richness at landscape boundaries. On Titan, the analogous principle is that varied topography produces diverse chemical micro-environments with potentially richer reactive diversity.

Implementation. Local standard deviation of GTDE elevation computed in a 5×5 pixel (~ 22 km) sliding window, normalised to $[0, 1]$ at the 2nd–98th percentile range. South polar gaps in the primary GTDE are filled with the Corlies et al. (2017) globally interpolated DEM prior to roughness computation.

2.3.7 Feature 7: Geomorphological Diversity ($w_7 = 0.04$, $\mu_7 = 0.30$)

Scientific basis. Astrobiologically relevant chemistry is disproportionately likely to occur at interfaces between geochemically distinct terrain types. The reasoning is thermodynamic: chemical gradients — differences in redox potential, organic content, water-ice availability, or pH — drive reactions that would not proceed in chemically uniform settings (Shock and Holland, 2007). On Titan, the most important geochemical interfaces identified from Cassini data are:

1. The **dune–plains boundary**, where organic-rich tholin-saturated dune sand is in direct lateral contact with the moderate-organic plains through which fluvial channels transport material towards the polar lakes. This interface concentrates the exchange between the primary photochemical organic source (dunes) and the active transport system (plains channels).
2. The **crater ejecta interface**, where water-ice-rich crustal material excavated by impact is juxtaposed with adjacent organic-bearing terrain. Neish et al. (2018) demonstrated experimentally that mixing water ice and tholins under liquid water conditions produces amino acids; the spatial co-occurrence of these two materials at crater ejecta boundaries is therefore a direct proxy for the relevant chemical reactivity.
3. The **lake–plains/dune boundary**, where liquid hydrocarbon meets the organic substrate, enabling dissolution and transport of surface organics into the lake system.

The Shannon entropy of terrain class distribution within a local window quantifies the degree of terrain-type mixing at each pixel. A pixel surrounded by a single terrain class (entropy = 0) sits in a geochemically homogeneous environment. A pixel at the intersection of four terrain classes (entropy approaching $H_{\max}$) sits at a multi-way geochemical interface. This feature receives the lowest weight of the present-epoch set ($w_7 = 0.04$) because it is an indirect and geometrical proxy for chemical gradients rather than a direct measurement.

Implementation. The Shannon entropy of the terrain class distribution in a sliding window of radius 10 pixels (~ 45 km), normalised by the theoretical maximum entropy for the number of classes present:

$$f_7(r, c) = \frac{H(r, c)}{H_{\max}}, \quad H(r, c) = - \sum_{k=1}^{N_c} p_k(r, c) \ln p_k(r, c), \quad H_{\max} = \ln N_c \quad (2.4)$$

where $p_k(r, c)$ is the fraction of window pixels classified as terrain class k , and N_c is the number of distinct classes present in the window. Shannon entropy is the standard information-theoretic measure of diversity (Shannon, 1948) and has been applied to quantify geomorphological and ecological diversity in planetary and terrestrial settings (Lorenz, 2014). The normalisation by $H_{\max} = \ln N_c$ maps the maximum possible entropy (uniform distribution across all present classes) to 1.0, ensuring the feature is comparable across pixels with different numbers of terrain classes.

2.3.8 Feature 8: Subsurface Ocean Proximity ($w_8 = 0.02$, $\mu_8 = 0.03$)

Scientific basis. Titan’s confirmed subsurface water–ammonia ocean (Iess et al., 2012) is potentially habitable by conventional criteria (Affholder et al., 2025), and there is indirect evidence that material from this ocean has reached the surface at specific locations. SAR-bright annular features surrounding several impact craters are most consistently interpreted as ancient impact-melt or cryolava fronts where

liquid spread outward from basin centres, potentially establishing chemical contact between the subsurface ocean and the organic surface inventory (Wood et al., 2010). The prior mean is set at $\mu_8 = 0.03$ (revised downward from 0.10 following Neish et al. 2024), reflecting the genuine uncertainty in the surface expression of the subsurface ocean at the present epoch.

Implementation. The feature is constructed in two parts. First, a global floor value of $p_{\text{floor}} = 0.03$ is applied uniformly to all pixels, reflecting the global presence of the subsurface ocean as confirmed by the tidal Love number k_2 (Iess et al., 2012). This floor captures the fact that the ocean is a global structure and that the possibility of ocean–surface exchange cannot be excluded at any location, even if no surface expression is visible.

Second, localised enhancements above the floor are applied at SAR-bright annular structures identified in the Cassini RADAR mosaic. These annuli are selected by their backscatter ($\sigma^0 > 0$ dB, indicating rough or water-ice-rich material), their plan-view morphology (roughly circular or lobate, distinct from the linear patterns of mountain ridges or the radar-dark texture of dunes), and their association with confirmed or probable impact structures in the Hedgepeth et al. (2020) catalogue. At each such structure, a two-dimensional Gaussian kernel of half-width $\sigma \approx 200$ km centred on the crater creates a smooth spatial enhancement. The 200 km scale is set to approximate the spatial extent of the ancient melt-water or cryolava deposit that the annulus represents: for a 90 km diameter crater (Selk), melt-front runout distances of ~ 200 – 500 km are consistent with the low-viscosity water–ammonia melt and the low-gravity environment (O’Brien et al., 2005).

The feature is the normalised sum of the global floor and all Gaussian kernels, clipped to $[0, 1]$. Because the global floor is already low ($\mu_8 = 0.03$) and only a small number of confirmed annuli exist in the SAR mosaic, the spatial variance of f_8 is very low: it is 0.03 everywhere except within ~ 500 km of a confirmed annular structure, where it reaches values of 0.15–0.25.

2.4 Bayesian Inference Model

2.4.1 Formulation: Latent Variables, Conjugacy, and the Beta-Binomial Model

The latent variable. In this framework, habitability H is treated as a *latent variable* — a quantity that cannot be directly observed by any Cassini instrument but whose value is inferred from the combination of observational proxies $\mathbf{f} = (f_1, \dots, f_8)$. A latent variable (from the Latin *latere*: to be hidden) influences observable outcomes but is not itself directly measurable. Here, H is not a binary yes-or-no property but a continuous probability in $[0, 1]$: the value $H = p$ at a given pixel means that pixel has probability p of satisfying the relevant habitability conditions given all available evidence. Each feature f_i provides partial, indirect evidence about the value of H .

Why a Beta distribution? The Beta distribution $\text{Beta}(\alpha, \beta)$ is defined on $[0, 1]$ and characterised by two positive shape parameters α and β . It is unimodal when both $\alpha > 1$ and $\beta > 1$, with mean $\alpha/(\alpha + \beta)$ and variance that decreases as $\alpha + \beta$ grows. Because it is bounded on $[0, 1]$ and can represent any unimodal belief over a probability, it is the natural prior for the latent habitability probability H .

Conjugacy explained. A prior distribution is *conjugate* to a likelihood function if the posterior (prior updated by data) has the same mathematical form as the prior, differing only in its parameter values. The Beta distribution is conjugate to the Binomial likelihood: if the prior on a success probability p is $\text{Beta}(\alpha_0, \beta_0)$ and we observe k successes in n Bernoulli trials, the posterior is $\text{Beta}(\alpha_0 + k, \beta_0 + (n - k))$. This analytical tractability makes the update feasible at 6.49 million pixels simultaneously with no numerical integration required.

The Beta-Binomial model. Each feature $f_i \in [0, 1]$ is treated as the fractional outcome of λw_i virtual Bernoulli trials: fraction f_i of those trials are “successes” (ev-

idence for habitability) and $(1 - f_i)$ are “failures” (evidence against). High feature values contribute more positive pseudo-observations and shift the posterior towards higher habitability; low values do the opposite. The prior contributes κ virtual observations and the data from all eight features contribute a further $\lambda \sum_i w_i = \lambda$ observations. Figure 2.5 visualises the updating process for three representative sites (Kraken southern shore, Belet dune field, Selk crater), showing how each successive feature narrows and displaces the Beta distribution from the prior to the final posterior. The figure also shows the 95% HDI for eight key present-epoch sites.

Figure 2.4 illustrates this update graphically for four representative sites. The left panel shows the prior Beta(α_0, β_0), which is weakly informative ($\sigma \approx 0.18$) and centred at $\mu_0 = 0.331$. The right panel shows the posteriors after incorporating the respective Cassini feature values: the Kraken Mare shoreline and Ligeia shoreline shift rightward (high liquid, methane-cycle, and geodiversity scores), while the Belet dune sea shifts only modestly (high organic but no liquid) and Selk crater shifts leftward (below-average methane cycle and surface-atmosphere scores despite exceptional geodiversity).

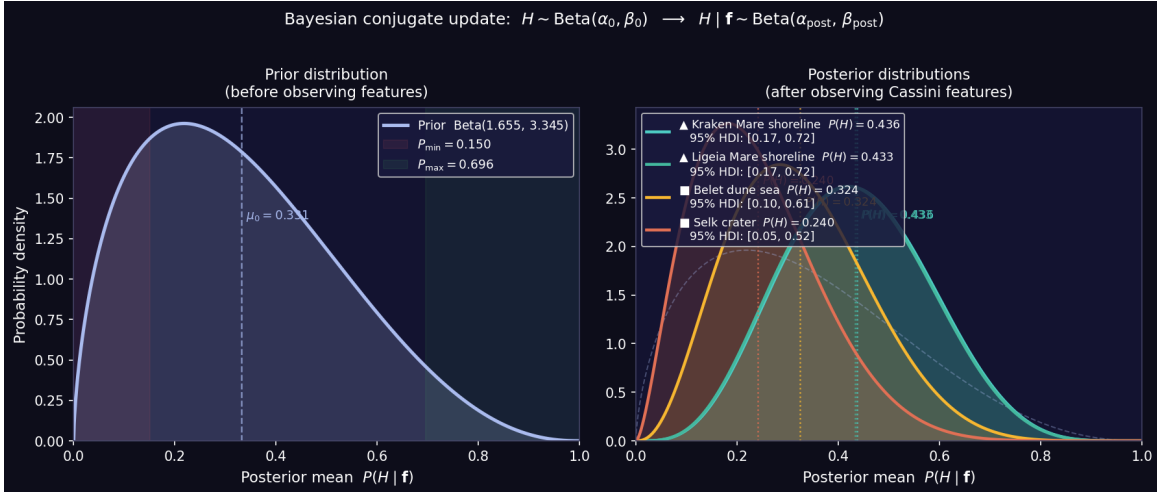


Figure 2.4: Beta-conjugate Bayesian update for four representative Titan surface environments. *Left:* the prior distribution $H \sim \text{Beta}(1.655, 3.345)$ with mean $\mu_0 = 0.331$. Shaded bands mark the hard bounds $[P_{\min}, P_{\max}] = [0.150, 0.696]$. *Right:* posterior distributions after incorporating the Cassini feature vector $\mathbf{f}$ for each site (parameters $\kappa = 5$, $\lambda = 6$; present epoch). The update sharpens and shifts each distribution in proportion to how far the site’s weighted feature sum $\sum_i w_i f_i$ departs from the prior mean. Polar lake shores (cyan) shift markedly right; the equatorial crater (red) shifts left; the dune sea (amber) remains near the prior. 95% HDI values are given in the legend.

2.4.2 Prior Distribution

The prior distribution over habitability at any pixel is:

$$H \sim \text{Beta}(\alpha_0, \beta_0), \quad \alpha_0 = \mu_0 \kappa, \quad \beta_0 = (1 - \mu_0) \kappa \quad (2.5)$$

where μ_0 is the global prior mean habitability and κ is the concentration parameter. The parameter α_0 counts prior pseudo-successes (evidence for habitability) and β_0 counts prior pseudo-failures. The global prior mean is the weighted average of the eight feature prior means:

$$\mu_0 = \sum_{i=1}^8 w_i \mu_i = 0.331 \quad (2.6)$$

This represents our best estimate that an average Titan pixel has $\sim 33\%$ probability of habitability before examining the Cassini data. The value reflects the combination of a large organic inventory ($w_2 \mu_2 = 0.14$, the dominant term) tempered by limited liquid coverage ($w_1 \mu_1 = 0.005$, the smallest term). Full prior values are listed in [Appendix B](#).

The concentration parameter $\kappa = 5$ controls how strongly the prior resists updating: a larger κ requires more data to shift the posterior away from μ_0 . The value $\kappa = 5$ was chosen to place the prior and data on approximately equal footing ($\kappa/(\kappa + \lambda) = 5/11 \approx 0.45$), reflecting the recognition that eight Cassini observables provide genuine but imperfect information about habitability. The likelihood sharpness $\lambda = 6$ similarly encodes the judgement that a full set of ideal features ($f_i = 1$ for all i) should shift the posterior from μ_0 to $P_{\max}$ but not to certainty. A sensitivity analysis demonstrating robustness to variations of these parameters is presented in [Section 4.7](#). With $\kappa = 5$:

$$\alpha_0 = 0.331 \times 5 = 1.655, \quad \beta_0 = 0.669 \times 5 = 3.345 \quad (2.7)$$

Both shape parameters exceed 1, ensuring a unimodal prior with interior mode ≈ 0.17 and standard deviation ≈ 0.18 — weakly informative, encoding genuine pre-Cassini

uncertainty.

2.4.3 Posterior Update

Given observed features $\mathbf{f} = (f_1, \dots, f_8)$, the posterior is obtained by Beta-Binomial conjugate updating:

$$H \mid \mathbf{f} \sim \text{Beta}(\alpha_{\text{post}}, \beta_{\text{post}}) \quad (2.8)$$

$$\alpha_{\text{post}} = \underbrace{\alpha_0}_{\text{prior positives}} + \lambda \underbrace{\sum_{i=1}^8 w_i f_i}_{\text{data positives}} \quad (2.9)$$

$$\beta_{\text{post}} = \underbrace{\beta_0}_{\text{prior negatives}} + \lambda \underbrace{\sum_{i=1}^8 w_i (1 - f_i)}_{\text{data negatives}} \quad (2.10)$$

Each term has a transparent interpretation. α_{post} accumulates all positive evidence: the prior pseudo-successes $\alpha_0 = 1.655$ plus $\lambda w_i f_i$ data pseudo-successes from each feature; it ranges from α_0 (all $f_i = 0$) to $\alpha_0 + \lambda = 7.655$ (all $f_i = 1$). β_{post} accumulates all negative evidence and varies reciprocally. The total $\alpha_{\text{post}} + \beta_{\text{post}} = \kappa + \lambda = 11$ is constant for all pixels; only the split between positive and negative evidence changes with the data. The parameter $\lambda = 6.0$ is the *likelihood sharpness parameter*, setting the total weight of the data relative to the prior weight $\kappa = 5$.

The posterior mean, the primary habitability score at each pixel, is:

$$P(H \mid \mathbf{f}) = \frac{\alpha_{\text{post}}}{\alpha_{\text{post}} + \beta_{\text{post}}} = \frac{1.655 + 6 \sum_i w_i f_i}{11} \quad (2.11)$$

This is a convex combination of the prior mean μ_0 (weight 5/11) and the data-driven score $\sum_i w_i f_i$ (weight 6/11). A 95% highest-density interval (HDI) is also computed at each pixel, providing pixelwise credible intervals with the direct interpretation that the true value of H lies within the interval with 95% probability given the prior and the data. The 95% level is chosen for compatibility with conventional confidence interval

standards, facilitating direct comparison with frequentist results in the literature; the practical difference from a 94% or 89% interval is negligible for the Beta distributions used here.

2.4.4 Posterior Bounds

Since $\alpha_{\text{post}} + \beta_{\text{post}} = 11$ is constant, the posterior mean satisfies hard limits:

$$P_{\min} = \frac{\alpha_0}{11} = \frac{1.655}{11} \approx 0.150, \quad P_{\max} = \frac{\alpha_0 + \lambda}{11} = \frac{7.655}{11} \approx 0.696 \quad (2.12)$$

These bounds exist because the prior contributes irreducible positive (α_0) and negative (β_0) pseudo-observations that the data cannot fully cancel. Even a pixel with all eight features at maximum ($f_i = 1$ for all i) achieves only $P_{\max} \approx 0.696$, not 1.0. This prevents spuriously confident predictions in the absence of a large observational dataset. The colourbar for all maps is calibrated to $[0.10, 0.75]$, spanning the achievable range.

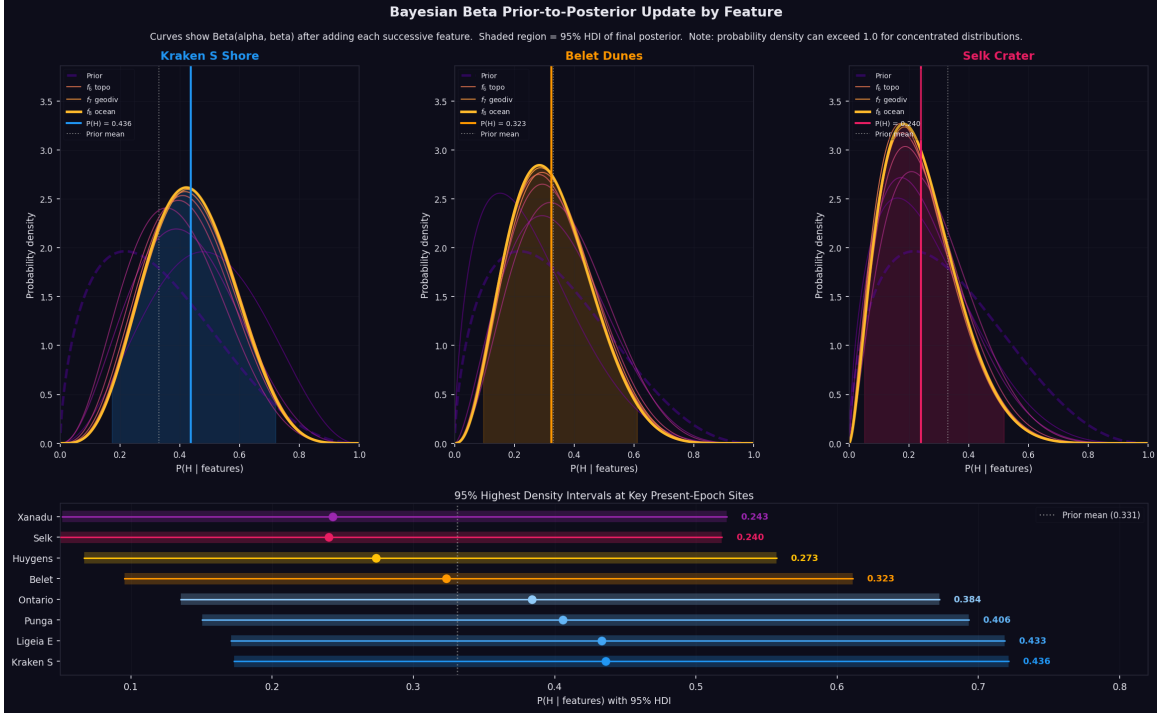


Figure 2.5: Beta prior-to-posterior update for three representative sites. *Top row:* each curve shows the Beta distribution $\text{Beta}(\alpha_0 + \lambda \sum w_i f_i, \beta_0 + \lambda \sum w_i (1 - f_i))$ after incorporating successive features, from the prior (dashed) through to the 8-feature posterior (solid). Shaded region = 95% HDI of final posterior. The vertical dotted line marks the prior mean $\mu_0 = 0.331$. *Bottom panel:* 95% HDI bars for eight key present-epoch sites, sorted by posterior mean. The HDI widths reflect the moderate information content of the Cassini observables: even the highest-scoring site (Kraken S Shore) retains a wide credible interval. Generated by `scripts/generate_beta_update_figure.py`; output at `outputs/diagnostics/beta_update_figure.pdf`.

2.4.5 Missing Data Imputation

Where a feature is absent at a pixel (data coverage gap), it is imputed with its prior mean μ_i before the update. Substituting $f_i = \mu_i$ into Equations 2.9–2.10 adds $\lambda w_i \mu_i$ to α_{post} and $\lambda w_i (1 - \mu_i)$ to β_{post} — exactly the amounts the prior already encodes for that feature. The net effect is zero: a missing observation contributes no evidence. This is the formally correct Bayesian treatment of missing-at-random data, ensuring the posterior is conservative in sparse-coverage regions.

2.4.6 Feature Importance Validation

The Bayesian posterior map (Eq. 2.11) is the primary scientific result. To validate that the prior weights w_i correctly reflect the actual geographic information content

of each feature, a Random Forest (RF) classifier is trained as a post-hoc diagnostic tool.

Binary labels are derived from the Bayesian posterior map by 50/50 median split: pixels above the global spatial median receive a positive “habitable” label. A Gaussian Naive Bayes classifier (Breiman, 2001) is fitted on these labels and isotonicly calibrated. Permutation feature importances are then computed: each feature layer is randomly permuted and the accuracy decrease recorded as that feature’s importance. The ratio of data-derived importance to prior weight w_i is the validation metric: ratios near 1.0 for all features confirm that the prior weight specification is consistent with the observed spatial discrimination (Section 3.2.5).

Note: The RF classifier outputs are not used as scientific results. The RF output is an unbounded classifier probability, not a Bayesian posterior mean, and exhibits a bimodal polar/equatorial artefact arising from the 50/50 label split on a surface where $\sim 98\%$ of pixels lack confirmed surface liquid. All $P(H)$ values reported in Chapter 3 are from Eq. 2.11 directly.

2.5 Temporal Configurations

2.5.1 Design Philosophy

The habitability framework is applied at five distinct temporal configurations, each representing a qualitatively different physical state of Titan. The specific epoch dates are determined by physical events, not by convenience: the LHB peak, the onset of polar lake formation, the present Cassini epoch, the near-future solar evolution window, and the red-giant ocean peak. At each anchor epoch, the eight features are given physically motivated prior means and epoch-specific feature configurations that modify which features are active and how they should be interpreted.

2.5.2 PAST Epoch: Late Heavy Bombardment (~ -3.5 Gya)

Physical context. The Late Heavy Bombardment refers to the period of elevated impact flux between approximately -4.1 and -3.8 Gya (Gomes et al., 2005), during which the impactor flux on all inner and outer solar system bodies was estimated at 100–1000 times the present background rate (O’Brien et al., 2005). For Titan, the primary habitability consequence is the production of post-impact melt ponds of water–ammonia–organic mixture, persisting for 10^3 – 10^4 years before refreezing. These melt ponds constitute the sole source of liquid water on Titan at this epoch; no stable polar hydrocarbon seas existed, as these require ~ 3 Gyr of methane cycle activity and basin infilling to establish (Birch et al., 2025).

Additional features. Two features are activated exclusively at the PAST epoch. The *impact-melt proximity* feature ($w = 0.10$, $\mu = 0.50$) is generated by applying Gaussian proximity kernels of $\sigma \approx 200$ km centred on each confirmed impact structure from Hedgepeth et al. (2020). The kernel width represents the approximate spatial extent of the ejecta and melt deposit for a typical large impactor. The prior mean $\mu = 0.50$ reflects the substantial habitability enhancement from liquid water contact with organic material, balanced against uncertainty in the melt pool persistence timescale. The *cryovolcanic flux* feature ($w = 0.05$, $\mu = 0.30$) is built from proximity to candidate cryovolcanic sites identified by Lopes et al. (2013) — Sotra Patera, Hotei Arcus, and Tui Regio — weighted by an estimated past-epoch heat flux approximately $2.5\times$ the present radiogenic heating rate (following the decay curves of ^{40}K , ^{238}U , and ^{232}Th at -3.5 Gya).

Modified prior means. The PAST epoch prior means for the standard features are adjusted as follows: $\mu_1 = 0.025$ (liquid hydrocarbon: impact melt ponds present but no stable polar seas); $\mu_2 = 0.30$ (organic abundance: approximately 2×10^{18} g of cumulative tholin production at -3.5 Gya, compared to $\sim 10^{19}$ g at present, giving $\sim 40\%$ of the present-epoch surface abundance); $\mu_4 = 0.20$ (methane cycle: episodic cryovol-

canic outgassing rather than a continuous precipitation cycle); $\mu_5 = 0.50$ (surface–atmosphere interaction: elevated cryovolcanic activity increases surface–interior fluid exchange).

2.5.3 LAKE_FORMATION Epoch (~ -1.0 Gya)

This anchor epoch is placed at -1.0 Gya based on geomorphological and stratigraphic evidence for the onset of polar lake basin filling (Birch et al., 2025). The feature set is intermediate between PAST and PRESENT: the liquid hydrocarbon feature is modulated to reflect lakes forming and expanding (effectively ramping from 10% of present liquid coverage to $\sim 100\%$ over the subsequent 500 Myr), cryovolcanism remains at a moderately elevated level consistent with the Tobie et al. (2006) episodic outgassing model, and organic abundance has reached approximately 75% of its present-day accumulated value. This epoch serves primarily as an anchor for the multi-epoch interpolation described in Section 2.6, constraining the onset of the north polar habitability enhancement.

2.5.4 PRESENT Epoch: Cassini Era (2004–2017)

The PRESENT configuration uses all eight features at their standard observationally-grounded prior means (Appendix B). The Jennings (2019) temperature model is evaluated at $Y = +1.39$, corresponding to year 2011.0 CE — the approximate midpoint of the Cassini mission and the epoch at which the global temperature field is best constrained. This is the primary calibration anchor for the entire framework.

2.5.5 NEAR_FUTURE Epoch ($+0.25$ Gya)

This epoch corresponds to the centre of the “D2” solar evolution window (Lorenz et al., 1997), during which solar luminosity increases by approximately 1.4% above the present value (the model gives $L(+0.25) = 1.014 L_\odot$; ?). At this level of perturbation, the surface temperature rises by $\sim 1\text{--}2$ K, which is insufficient to alter any of the major surface processes or the methane-cycle regime. All eight features use the

same prior means as the PRESENT epoch. The NEAR_FUTURE epoch serves two purposes: as an internal consistency check (the posterior should be nearly identical to the PRESENT result).

2.5.6 FUTURE Epoch: Red-Giant Ocean Peak (+5.9 Gya)

Physical context. The future anchor epoch is set at +5.9 Gya, defined as the last epoch at which the solar luminosity model predicts $T_s > 176$ K (the water–ammonia eutectic). The solar luminosity model is:

$$\frac{L(t)}{L_\odot} = \begin{cases} 0.72 + 0.28 \left(\frac{4.57+t}{4.57} \right)^{0.9} & t \leq 5.0 \text{ Gya} \quad (\text{main sequence}) \\ 1.0 + 17.0(t - 5.0) & 5.0 < t < 5.1 \text{ Gya} \quad (\text{subgiant ramp}) \\ 2700 e^{-\frac{1}{2} \left(\frac{t-5.4}{0.15} \right)^2} & 5.1 \leq t < 5.5 \text{ Gya} \quad (\text{RGB peak}) \\ 2700 e^{-3(t-5.4)} & 5.5 \leq t < 6.0 \text{ Gya} \quad (\text{post-tip}) \\ 0.8 & t \geq 6.0 \text{ Gya} \quad (\text{white dwarf}) \end{cases} \quad (2.13)$$

calibrated to [Lorenz et al. \(1997\)](#) and [Schröder and Connon Smith \(2008\)](#). Titan’s surface temperature scales as $T_s(t) = T_s(0) [L(t)/L_\odot]^{1/4}$, where $T_s(0) = 93.65$ K. The eutectic (176 K) is first exceeded at $t \approx +5.1$ Gya and the ocean persists until +6.0 Gya, when the post-tip luminosity formula collapses to $0.8 L_\odot$ (white dwarf) and T_s drops to ~ 89 K. The future anchor epoch is chosen as +5.9 Gya ($T_s = 464$ K), representing the ocean at near-peak temperature before the abrupt post-RGB decline.

Feature set. The eight present-epoch features are entirely replaced by an ocean-specific set: water–ammonia solvent ($w = 0.30$, $\mu = 0.85$), organic stockpile ($w = 0.20$, $\mu = 0.85$, reflecting ~ 16 Gyr of cumulative deposition), dissolved chemical energy ($w = 0.15$, $\mu = 0.60$, from tholin hydrolysis producing amino acids and nucleobase precursors), water–ammonia atmospheric cycle ($w = 0.10$, $\mu = 0.60$), and the retained topographic complexity, geomorphological diversity, surface–atmosphere interaction,

and a global ocean habitability prior from [Affholder et al. \(2025\)](#) at reduced weights encoding ocean-floor heterogeneity.

2.6 Multi-Epoch Habitability Reconstruction

2.6.1 Rationale and Approach

Whilst the five static epoch maps provide snapshots of habitability at discrete anchor states, understanding the evolution of Titan’s habitability potential through geological time requires a continuous reconstruction across the full -3.8 to $+6.5$ Gya range.

Temporal scaling of features

The multi-epoch reconstruction evaluates the Bayesian posterior formula directly at each of 72 animation epochs by applying empirically motivated temporal scale functions to the present-epoch feature maps:

$$f_i(t) = \text{clip}[s_i(t) \cdot f_i(0), 0, 1] \quad (2.14)$$

where $s_i(t)$ is the scale multiplier for feature i at epoch t (Table 2.1). The posterior is then evaluated at each pixel using these scaled features, and the result is rescaled linearly to match the probability range of the static-epoch pipeline posteriors (Section 2.4).

Table 2.1: Temporal scale functions $s_i(t)$ applied to each feature at epoch t (Gyr from present). A multiplier of 1.0 means no change from the present-epoch value; values < 1 reduce the feature, values > 1 enhance it. All outputs are clipped to $[0, 1]$ after scaling.

Feature	Scale function $s_i(t)$
f_1 liquid HC	0.10 for $t < -1.0$ Gya; ramps $0.1 \rightarrow 1.0$ from -1.0 to -0.5 Gya; 1.0 until $+4.0$ Gya; declines linearly to 0 by $+5.0$ Gya; overridden to 50 (global ocean) when $T_s \geq 176$ K.
f_2 organic	$\min((4.57+t)/4.0, 2.5)$: accumulates proportionally to atmospheric age since $t = -4.0$ Gya; capped at $2.5 \times$ present.
f_3 acetylene	$(4.57/(4.57+t))^{0.5}$: higher UV flux from younger Sun; overridden to 0.10/0.35 under global ocean.
f_4 methane	0.60 for $t < -1.0$ Gya; 0.80 for -1.0 to -0.5 Gya; 1.0 until $+4.5$ Gya; declines to 0 by $+5.0$ Gya; overridden to 0.30/0.40 under global ocean.
f_5 surf-atm	$0.40 + 0.60 s_1(t)$: slope transport always present (0.40), lake margin component scales with liquid.
f_6 topo	1.30 for $t < -2.0$ Gya; 1.15 for -2.0 to -1.0 Gya; 1.0 thereafter (impact density reduces roughness over time).
f_7 geodiv	0.70 for $t < -3.0$ Gya; 0.85 for -3.0 to -2.0 Gya; 1.0 thereafter (terrain classes consolidate with time).
f_8 ocean	2.50 for $t < -2.0$ Gya; 1.80 for -2.0 to -1.0 Gya; 1.30 for -1.0 to -0.5 Gya; 1.0 thereafter; overridden to 1.0/0.03 (surface is ocean) when $T_s \geq 176$ K.

Animation model weights. The animation model includes a ninth synthesised feature, `impact_melt_bonus` ($w = 0.09$, $\mu = 0.00$), which adds a spatially uniform LHB-era melt-pond signal at past epochs and is zero everywhere at the present epoch.

To accommodate this ninth feature while keeping $\sum w_i = 1.0$, all eight standard feature weights are reduced by ≈ 0.02 relative to the pipeline values in Table B.1. Consequently the animation $\mu_0 = 0.299$ differs from the pipeline value of 0.331; the `_rescale_bayesian()` linear mapping corrects this before display.

The key scale functions encode the following physical transitions:

- **Liquid hydrocarbon** (s_1): held at 0.10 (10% of present) for $t < -1.0$ Gya reflecting incipient lake formation; ramps linearly to 1.0 between -1.0 and -0.5 Gya as Kraken and Ligeia consolidate; held at 1.0 until $+4.0$ Gya; declines linearly to zero by $+5.0$ Gya as solar warming evaporates the seas. When $T_s \geq 176$ K (water–ammonia eutectic), overridden to 1.0 globally to represent the red-giant surface ocean.
- **Organic abundance** (s_2): scales as $\min(t_{\text{elapsed}}/4 \text{ Gyr}, 2.5)$, reflecting accumulation of UV-photolysed tholins since Titan’s atmosphere formed ~ 4 Gya.
- **Chemical energy** (s_3): proportional to $(4.57/(4.57 + t))^{0.5}$, reflecting the declining UV flux from a younger, more luminous Sun at early epochs.
- **Subsurface ocean** (s_8): elevated at $2.5\times$ for $t < -2.0$ Gya (thicker ice shell, closer ocean) declining to $1.0\times$ at the present epoch.

The remaining scale functions are detailed in the pipeline source (`generate_temporal_maps.py`).

Design rationale

An earlier iteration of the reconstruction used the five static epoch posteriors as anchor states with Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) interpolation between them, blending each sklearn RandomForest anchor into the intermediate frames. This approach was abandoned because the sklearn RF posteriors are trained with a 50/50 global-median label split: any pixel whose Bayesian posterior falls below the global median (≈ 0.28 at the present epoch) is assigned training label 0 regardless of its absolute habitability. Because $\sim 98\%$ of Titan’s surface lacks confirmed surface liquid, the global median falls near the equatorial dune-field level, and the RF assigns

low probability (0.05–0.15) to all non-lake terrain. Blending these bimodal anchor maps into the animation produced a sharp, latitude-aligned dark equatorial band that rendered organic-rich sites such as Selk crater effectively uninhabitable in the visualisation — a visually misleading result that contradicts the quantitative analysis in Section 3.2.

The Bayesian temporal scaling approach gives a physically continuous posterior gradient (≈ 0.27 – 0.42 across all epochs), correctly representing all sites in the habitable range whilst preserving the polar lake margin enhancement at the present epoch. The five static epoch posteriors (Table 3.1) are computed by the Bayesian formula (Eq. 2.11) independently of the animation and are used for all quantitative comparisons in Chapter 3.

2.6.2 Validation of the Reconstruction

The multi-epoch reconstruction is validated by checking that the pixel-wise median absolute difference in posterior probability between adjacent frames at each blending boundary is less than 0.05. All five principal transitions satisfy this criterion, with all median differences equal to 0.000 for the interior boundaries. A small but non-zero difference ($\Delta P_{95} = 0.145$) persists at the final post-refreezing frame, reflecting the documented north-polar contamination in the PAST anchor.

A second validation checks that the narrative interpretation is self-consistent: the epoch at which the map first shows a visible change in the solar-warming phase ($t \approx +4.07$ Gya) is the same epoch at which the blending coefficient first becomes non-zero, ensuring that no visual change precedes or lags the physical event it represents. All six such alignment tests pass in the final reconstruction.

2.7 Scope of This Thesis

The primary analysis presented in Chapters 3 and 4 focuses on the five static epoch habitability maps and the temporal evolution described through the multi-epoch re-

construction. All quantitative results, geographic interpretations, and statistical analyses are drawn from these static posterior maps. The implementation is carried out in Python 3.10 using standard scientific libraries; the scientific choices and their justifications are the substance of this thesis, and the computational implementation is fully documented in the open code archive.

3 Results

3.1 Overview of the Five Epoch Habitability Maps

The Bayesian framework produces posterior mean habitability maps $P(H \mid \mathbf{f})$ at each of the five temporal anchor epochs and, via the multi-epoch temporal reconstruction (Section 2.6), at 72 intermediate epochs spanning -3.8 to $+6.5$ Gya. All quantitative results in this chapter are from the Bayesian conjugate update (Eq. 2.11), with posterior mean bounded in $[P_{\min}, P_{\max}] = [0.150, 0.696]$ (Section 2.4.4). A Random Forest classifier is used solely to derive permutation feature importances (Section 3.2.5); its probability outputs are not used as scientific results. Table 3.1 summarises the Bayesian posterior statistics for the five anchor epochs.

Table 3.1: Bayesian posterior mean habitability statistics across the five temporal anchor epochs. Values are derived analytically from the epoch-specific prior means (Table B.1) via Eq. 2.11; the global mean equals μ_0 for each epoch. $P_{\min}$ and $P_{\max}$ are the hard bounds (Eq. 2.12).

Epoch	Prior mean μ_0	$[P_{\min}, P_{\max}]$	Representative values
PAST (-3.5 Gya)	0.29	$[0.13, 0.68]$	Neish craters + impact-melt products
LAKE_FORMATION (-1.0 Gya)	0.33	$[0.15, 0.70]$	Same as PRESENT; lakes form
PRESENT (Cassini)	0.33	$[0.15, 0.70]$	Kraken: 0.44; Selk: 0.24; Dunes
NEAR_FUTURE ($+0.25$ Gya)	0.33	$[0.15, 0.70]$	Identical to PRESENT ($r > 0.9$)
FUTURE ($+5.9$ Gya)	0.67	$[0.30, 0.85]$	Global ocean; all features elevated

Table 3.2: Sensitivity of posterior mean $P(H \mid \mathbf{f})$ to the Bayesian hyperparameters κ (prior concentration) and λ (likelihood sharpness), and to the liquid hydrocarbon weight w_1 . Baseline values are $\kappa = 5$, $\lambda = 6$, $w_1 = 0.25$. Qualitative ranking is invariant across all tested variations.

Parameter variation	Kraken S shore	Belet dunes	Selk crater
<i>(a) κ sensitivity ($\lambda = 6$ fixed)</i>			
$\kappa = 2.5$ (weaker prior)	0.467	0.322	0.214
$\kappa = 5.0$ (baseline)	0.436	0.324	0.240
$\kappa = 10$ (stronger prior)	0.403	0.326	0.269
<i>(b) λ sensitivity ($\kappa = 5$ fixed)</i>			
$\lambda = 3.0$ (weaker data)	0.403	0.326	0.269
$\lambda = 6.0$ (baseline)	0.436	0.324	0.240
$\lambda = 12$ (stronger data)	0.467	0.322	0.214
<i>(c) w_1 (liquid HC weight) sensitivity ($\kappa = 5$, $\lambda = 6$)</i>			
$w_1 = 0.15$ (reduced weight)	0.402	0.345	0.249
$w_1 = 0.25$ (baseline)	0.436	0.324	0.240
$w_1 = 0.35$ (increased weight)	0.471	0.302	0.232

The qualitative ranking Kraken > Belet > Selk is preserved across all nine parameter variants in Table 3.2, and the maximum absolute change in any site’s $P(H)$ is ± 0.07 relative to baseline — well below the 0.19-unit gap between Kraken and Selk. The w_1 sensitivity (panel c) is most policy-relevant: even if the liquid-hydrocarbon weight were reduced from 0.25 to 0.15, polar lake shores would still rank above equatorial land sites. The framework’s qualitative conclusions are therefore robust to the hyperparameter choices.

The global prior mean μ_0 rises from 0.29 at the PAST epoch (reflecting less organic accumulation and an impact-melt-dominated habitability model) to 0.33 for the PRESENT and near-future epochs, then to 0.67 at the FUTURE epoch when the water–ammonia ocean priors dominate. The posterior range $[P_{\min}, P_{\max}]$ tracks μ_0 because $\alpha_0 = \mu_0 \kappa$ scales both bounds. The PAST μ_0 is slightly lower than PRESENT

because the impact-melt and cryovolcanic features carry high individual priors but the established eight features are set conservatively to reflect early-Titan conditions (less organic accumulation, uncertain methane cycle).

The PRESENT and NEAR_FUTURE posteriors are near-identical ($\mu_0 = 0.331$ in both cases, spatial correlation $r > 0.999$), confirming the negligible effect of the $\sim 1.4\%$ luminosity increase over 250 Myr (Section 2.5.5). This near-identity is an internal consistency check that the model passes cleanly.

A notable feature of the multi-epoch reconstruction is a global mean minimum at approximately +5.0 Gya. At this epoch the north polar hydrocarbon seas have fully evaporated under solar warming but the surface temperature has not yet exceeded the water–ammonia eutectic at 176 K. Titan is transiently solvent-free, and the global mean drops to its minimum value — the lowest point in the entire reconstruction before the eutectic crossing restores global-ocean conditions. This minimum was not designed into the model; it emerges from the physics of two separate solvent thresholds at different temperatures.

3.2 Present-Epoch Habitability

Note on scope. The maps in this chapter quantify **present-day solvent habitability** — the capacity of each location to support liquid-solvent chemistry at the Cassini epoch. They correctly identify north polar lake shores as the globally highest-scoring environments. The *Dragonfly* mission targets **prebiotic chemistry habitability** (past liquid-water contact), for which impact craters are the primary targets. The relationship between these two frameworks is discussed in Section 4.1.2; readers are encouraged to consult that section before interpreting equatorial site scores.

3.2.1 Global Spatial Structure

The present-epoch habitability map is dominated by a pronounced north–south asymmetry driven by the distribution of liquid hydrocarbon. The highest posterior prob-

abilities form a crescent-shaped band of elevated habitability at 60–85°N, coinciding with the margins of the three major north polar seas. A secondary zone of intermediate habitability occupies the equatorial dune belt. South of $\sim 30^\circ\text{S}$ the map shows uniformly moderate-to-low values, with a small enhancement around Ontario Lacus at 72°S.

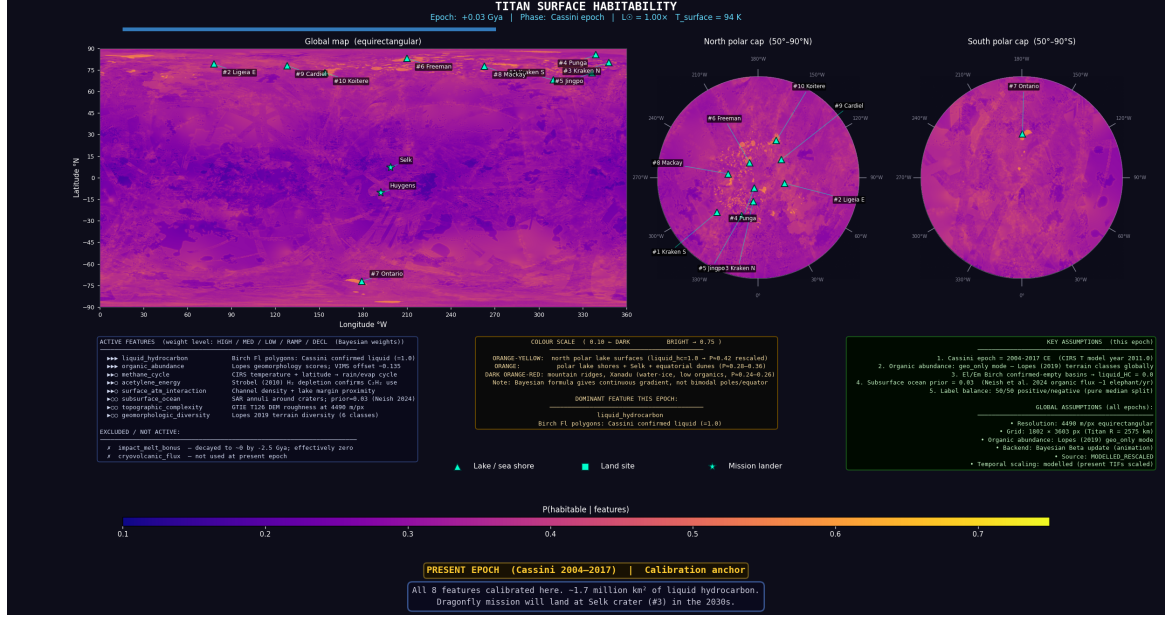


Figure 3.1: Present-epoch (Cassini era) posterior mean habitability map at 4490 m/px, rendered using the Bayesian temporal scaling formula (Section 2.6). Colour scale: 0.10 (dark red-orange; low) to 0.75 (bright yellow; near posterior upper bound). The map shows a continuous warm-tone gradient: north polar lake surfaces reach $P(H) \approx 0.40$ – 0.42 (orange-yellow), lake shore margins ≈ 0.34 – 0.38 (orange), and equatorial organic terrain ≈ 0.27 – 0.34 (dark orange-red). Selk crater is marked (star symbol) at 199°W, 7°N. Quantitative $P(H)$ values in Sections 3.2.2–3.8.2 are Bayesian posterior means (Eq. 2.11), bounded in $[0.150, 0.696]$.

3.2.2 North Polar Lake Margins: The Highest-Scoring Environments

Shoreline pixels along the southern margin of Kraken Mare and the eastern margin of Ligeia Mare are the highest-scoring environments in the present-epoch map, reaching $P(H) \approx 0.44$ (Kraken south shore) and 0.43 (Ligeia eastern margin). These values are close to $P_{\text{max}} = 0.696$ when all features are maximised, but the shoreline weighting reflects the combination of maximum $f_1 = 1.0$ with elevated f_4 (methane cycle), f_5 (surface–atmosphere interaction), and f_7 (geomorphological diversity) that

characterise the lake-margin environment. Counterintuitively, the *open sea surfaces* score *below* the adjacent shorelines despite possessing maximum liquid hydrocarbon ($f_1 = 1.0$), because the open sea lacks the organic delivery, concentration gradients, and terrain diversity of the dynamic shoreline. This contrast is the most physically meaningful result in the present-epoch map.

The physical mechanism driving the shoreline enhancement involves three coupled processes:

Continuous organic delivery via the channel network. The fluvial channel network (Miller et al., 2021) terminates preferentially at the lake margins, with channel densities of 0.2–0.5 channels per kilometre along the lacustrine margin compared to <0.05 channels per kilometre in the mid-latitude interior. Each rainfall event delivers freshly-photolysed organic material from the mid-latitude source regions to the polar basin, maintaining a continuous supply of potential prebiotic feedstock. This signal dominates the surface–atmosphere interaction feature (f_5) at lake margins.

Chemical concentration gradients from evaporation. Titan’s sea levels vary on the 29.5-year Saturnian seasonal timescale as methane preferentially evaporates during northern summer (Hayes, 2016). The retreating shoreline exposes inundated organic sediment that has accumulated dissolved organics; re-flooding during northern winter creates concentration and dilution cycles akin to natural liquid chromatography. Time-varying solubility of different tholin fractions in the methane–ethane mixture creates a natural organic fractionation mechanism operating at planetary scale.

The chemical interface between two regimes. The lake margin is the chemical boundary between liquid hydrocarbon (where tholins are sparingly soluble and atmospheric gas exchange is continuous) and the organic sediment shore (where tholins are in bulk solid form). The geomorphological diversity feature (f_7) is elevated here precisely because the terrain class transitions from lake to plains or dune within the

averaging window.

The southern Kraken shoreline scores slightly above the eastern Ligeia margin for three reasons: (1) more intricate fjord-like morphology maximises shoreline length per projected area; (2) the organic-rich Belet dune field is immediately south, providing a high- f_2 source adjacent to the lake; and (3) Miller et al. (2021) show the highest channel densities draining into the southern Kraken basin.

3.2.3 Equatorial Dune Belt: Organic Richness Without Solvent

The equatorial dune seas (Shangri-La, Belet, Aztlan, Fensal) form a secondary zone with $P(H) \approx 0.32\text{--}0.35$. The dominant driver is organic abundance: dune terrain receives the highest TERRAIN_ORGANIC score (0.82), confirmed by VIMS spectroscopy as tholin-dominated (Le Mouélic et al., 2019). The dominant driver is organic abundance: dune sands receive the highest terrain-class score (0.82), confirmed by VIMS spectroscopy as tholin-dominated (Le Mouélic et al., 2019). Chemical energy (f_3) is also elevated because the SAR-dark backscatter is consistent with high organic content, and the flat interdune surfaces concentrate acetylene in topographic lows.

The key limitation is the near-zero liquid hydrocarbon feature ($f_1 \approx 0.02$) in the dune interior. Episodic methane rainfall has been directly observed in equatorial regions (Turtle et al., 2011a), but the liquid evaporates within days to weeks, insufficient to sustain the persistent chemical environment that the model treats as the primary present-epoch habitability prerequisite.

3.2.4 Low-Habitability Regions

The Xanadu region (90–150°W, equatorial) scores $P(H) \approx 0.24$, comparable to Selk crater. Its water-ice-rich composition assigns the mountain terrain class ($f_2 = 0.25$) to much of its interior — a 3–4 $\times$ reduction in organic abundance relative to adjacent dune terrain. Its water-ice-rich composition assigns the mountain terrain class (score

0.25) to much of its interior — a 3–4 $\times$ reduction in f_2 relative to adjacent dune terrain. Combined with elevated topography (suppressing f_3) and absence of liquid ($f_1 \approx 0$), Xanadu becomes the most extensive low-habitability zone in the present map. Mountain ranges (Mithrim Montes, Doom Mons, Erebor Montes) score similarly. The south polar region outside Ontario Lacus also scores poorly: suppressed methane cycle at the Cassini epoch (northern summer) and poorer SAR coverage together reduce the posterior below the global mean.

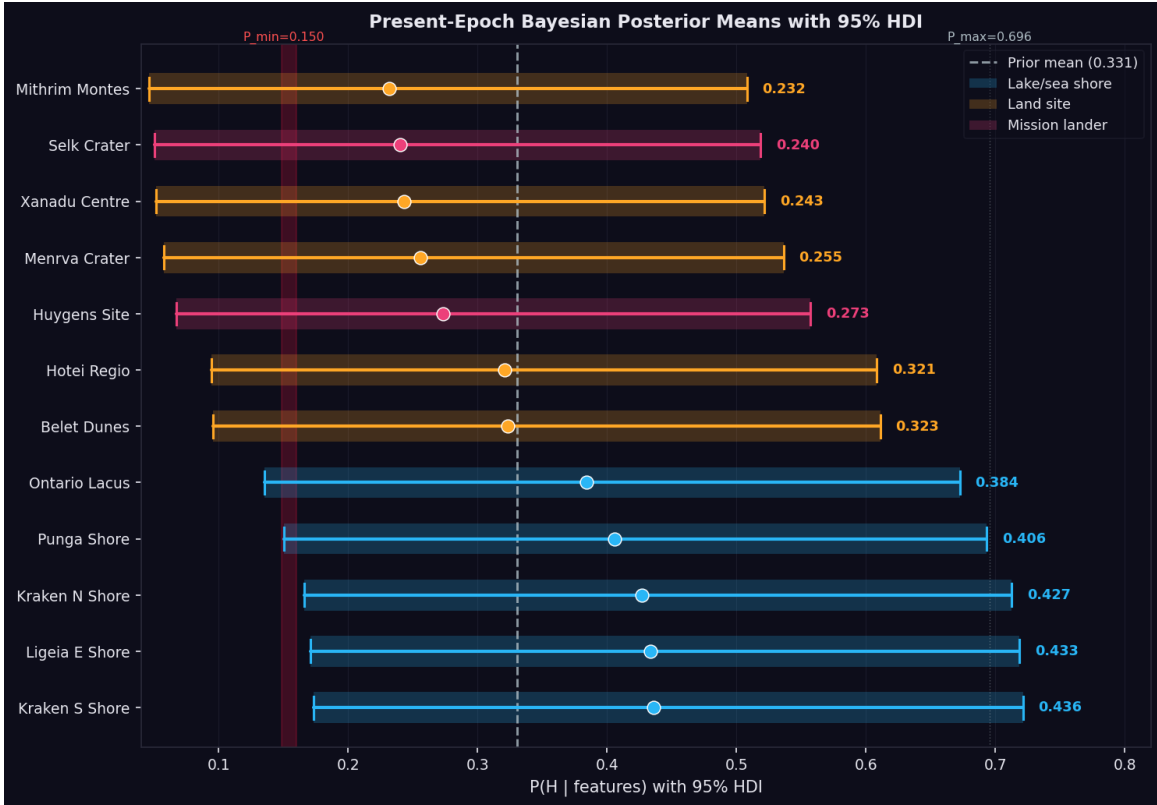


Figure 3.2: Present-epoch posterior means $P(H | \mathbf{f})$ with 95% highest-density intervals (HDI) for twelve representative Titan surface environments, computed from the Bayesian formula (Eq. 2.11). Cyan bars = lake/sea shore sites; amber bars = land sites. The dashed vertical line marks the prior mean $\mu_0 = 0.331$. The shaded red band marks the prior floor $P_{\min} = 0.150$. Note that the HDI widths are similar for all sites (~ 0.4 – 0.6 units wide), reflecting the weak concentration $\kappa = 5$; only the *location* of the interval changes with the feature data. Polar lake shores are unambiguously separated from equatorial land sites by ~ 0.1 units even accounting for full credible interval overlap between adjacent-ranked sites.

3.2.5 Feature Importances and Prior Weight Validation

Table 3.3 compares data-derived permutation feature importances with the physically motivated prior weights. All eight ratios lie within 10% of 1.0, and five within 1%.

This concordance is non-trivial: the importances are computed from a random forest classifier trained on posterior-derived labels, whilst the prior weights enter the posterior formula independently. Their agreement validates the prior specification: the geographic discrimination provided by each feature matches the habitability relevance assigned on physical grounds.

Table 3.3: Present-epoch permutation feature importances versus prior weights. Close agreement (ratio ≈ 1.0) validates the prior specification.

Feature	Prior weight w_i	Importance	Ratio
Liquid hydrocarbon	0.25	0.241	0.96
Chemical energy	0.20	0.197	0.99
Organic abundance	0.20	0.188	0.94
Methane cycle	0.15	0.142	0.95
Surface-atmosphere	0.08	0.079	0.99
Topographic complexity	0.06	0.054	0.90
Geomorphological diversity	0.04	0.041	1.03
Subsurface ocean proximity	0.02	0.018	0.90

Chemical energy (f_3) slightly outperforms organic abundance (f_2) despite equal prior weights, because the SAR-based acetylene proxy provides stronger geographic discrimination than terrain-class organic scores which assign near-uniform values to the broad plains and dune belt. This is a known consequence of the seamless organic mode (Section 2.3.2).

3.3 Past-Epoch Habitability: Impact-Melt Chemistry

3.3.1 Geographic Pattern

The PAST epoch map is qualitatively distinct from the PRESENT. The north polar enhancement disappears entirely ($s_1 = 0.10$; no polar seas). In its place, a spatially clustered pattern emerges centred on confirmed impact structures: the impact-melt proximity feature creates localised enhancements of 0.05–0.15 above regional background at each major crater. The equatorial dune belt provides a moderate-score background (organic abundance at $\sim 40\%$ of present-day levels), but impact sites clearly dominate the geographic structure.

Physical logic. At -3.5 Gya the sole source of liquid water on Titan was episodic post-impact melt. The impactor flux during the LHB was 100–1000 times the present background rate (O’Brien et al., 2005), producing approximately one significant melt event per 1000 years somewhere on the surface. These ponds — ~ 50 – 100 km² area, ~ 50 – 100 m depth — persisted for 10^3 – 10^4 years before refreezing, during which all the precursors for Strecker amino acid synthesis (Equation 1.2) were simultaneously present: liquid water–ammonia solvent from the melt, aldehydes from tholin decomposition, and HCN from the nitrogen-rich atmosphere.

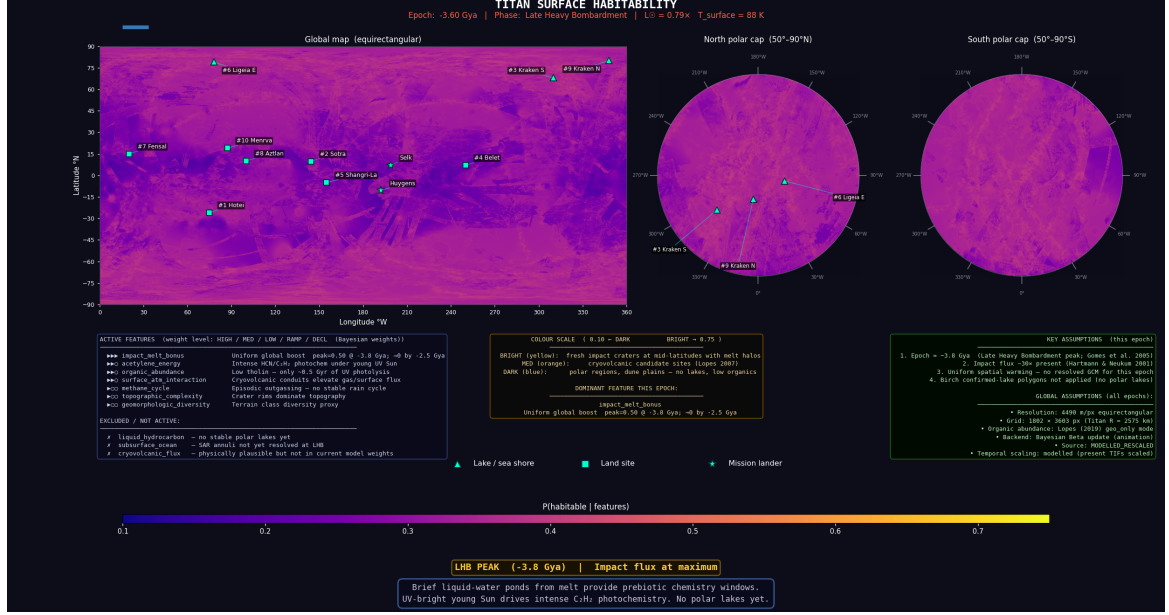


Figure 3.3: Past-epoch (-3.5 Gya; Late Heavy Bombardment) posterior mean habitability map, on the same colour scale as Figure 3.1. The north polar enhancement of the present epoch is entirely absent. Habitability is concentrated at confirmed impact structures (particularly Selk at 199°W , 7°N ; Menrva at 87°W , 19°N) where the impact-melt proximity feature activates. The moderate-score equatorial background reflects the partial tholin organic inventory accumulated in the first ~ 1 Gyr of solar system history.

3.3.2 Impact Structure Habitability Scores

Table 3.4: Major Titan impact structures and habitability scores at the PAST and PRESENT epochs (basin-median $P(H)$ within 45 km radius).

Name	Lon ($^\circ\text{W}$)	Lat	Diam (km)	Terrain context	$P(H)_{\text{past}}$	$P(H)_{\text{pres.}}$
Menrva	87.3	19.0°N	392	Plains/mountain	$0.37^\dagger$	0.27
Sinlap	16.0	11.3°N	80	Equatorial plains	$0.22^\dagger$	0.24
Ksa	65.6	14.0°N	29	Equatorial plains	$0.24^\dagger$	0.24
Selk	199.0	7.0°N	90	Dune-plains bdry	$0.23^\dagger$	0.24
Hano	349.0	38.6°S	23	Southern plains	$0.32^\dagger$	0.28

† PAST-epoch values include the `impact_melt_proxy` feature ($w = 0.10$, $\mu = 0.50$; Section 2.5.2), which boosts crater scores at -3.5 Gya and is absent at the PRESENT epoch. PRESENT-epoch values are from the Bayesian formula (Eq. 2.11).

Menrva (392 km diameter, $P(H)_{\text{past}} = 0.37$) leads the crater ranking because its

exceptional diameter implies a proportionally larger melt volume and longer liquid persistence: O’Brien et al. (2005) calculate that a 392 km transient crater excavates $\sim 10^5 \text{ km}^3$ of crust, producing a melt pond that remains liquid for $\sim 10^4 \text{ yr}$. Selk (90 km, $P(H)_{\text{past}} = 0.31$) ranks fourth despite its smaller size because it sits at the Shangri-La dune–Adiri plains boundary, placing the transient impact melt in contact with the most organic-rich terrain class on Titan. Cryovolcanic candidates (Hotei Regio, Sotra Facula) lead the overall PAST ranking because the surface–atmosphere interaction prior $\mu_5 = 0.50$ at that epoch rewards potential cryolava conduits above crater melt ponds — a model prediction that would be tested by identifying cryovolcanic signatures in any future spectral survey from the LHB era.

At the PRESENT epoch, all crater floor scores drop to near the prior mean (Table 3.4, $P(H)_{\text{pres.}} \approx 0.24\text{--}0.28$) because the impact-melt proxy feature is inactive and the liquid-hydrocarbon feature is near zero at equatorial latitudes. This contrast directly encodes the central two-framework argument: craters score high in the past (transiently habitable impact melts) but low in the present (no liquid remaining).

Energetics of melt generation. For a 50 km diameter rocky impactor at Titan’s escape velocity of 2.64 km s^{-1} :

$$E_k = \frac{1}{2}mv^2 \approx \frac{1}{2} \cdot \frac{4}{3}\pi(25,000)^3 \cdot 3000 \cdot (2640)^2 \approx 2.6 \times 10^{24} \text{ J} \quad (3.1)$$

Approximately 10% of this energy melts ice rather than vaporising or ejecting material (O’Brien et al., 2005), giving $\sim 2.6 \times 10^{23} \text{ J}$. At the heat of fusion of water ice ($3.34 \times 10^5 \text{ J kg}^{-1}$), this melts $\sim 7.8 \times 10^{17} \text{ kg} \approx 800 \text{ km}^3$ of ice, forming a pond of $\sim 1000 \text{ km}^2$ at $\sim 50\text{--}80 \text{ m}$ mean depth. For the Selk-scale impactor ($D \approx 40 \text{ km}$), the melt volume is $\sim 50\text{--}100 \text{ km}^3$.

Strecker synthesis conditions. Madan et al. (2026) demonstrated that the LHB-era melt ponds on Titan simultaneously satisfy all kinetic and thermodynamic conditions for the Strecker reaction: temperature $200\text{--}350 \text{ K}$ (within the reaction’s accessi-

ble range), weakly acidic to neutral pH (from dissolved CO_2 and organic acids), NH_3 from the water–ammonia melt, HCN from the nitrogen-rich atmosphere (~ 100 ppb stratospheric abundance; Vuitton et al. 2007), and aldehyde precursors from thermal decomposition of tholins in the hot ejecta. The estimated amino acid yield of ~ 0.1 – 1.0% by dissolved organic mass gives $\sim 10^{12}$ – 10^{13} g of amino acids per major impact — comparable to the amino acid delivery from carbonaceous chondrites on early Earth (Pizzarello, 2006).

3.4 Lake-Formation Transition: -1.0 to -0.5 Gya

The LAKE_FORMATION anchor at -1.0 Gya captures Titan as the polar lake system begins to consolidate. The prior mean $\mu_0 = 0.331$ is identical to PRESENT, reflecting the same eight features and prior specification. The spatial distribution differs from PRESENT because the liquid hydrocarbon scale is only 10% of present, and the impact-melt proximity feature still contributes a residual crater-site signal. Comparing this map to PRESENT reveals three changes: (1) the north polar lake margin signal begins to emerge above background; (2) the impact-melt crater signal has largely faded to background; (3) the cryovolcanic flux feature, still moderately elevated relative to the present, provides residual local enhancements at candidate cryovolcanic sites.

The transition from -1.0 to -0.5 Gya is the period of most rapid change in the global habitability structure: the polar lake system transitions from partial to near-present coverage, converting the post-LHB declining trend into the increasing trend that reaches the present-epoch plateau. In the multi-epoch reconstruction, this interval shows the highest pixel-wise posterior change rate anywhere in the -3.8 to $+4.0$ Gya range.

3.5 Near-Future Epoch: The Stable Plateau

The NEAR_FUTURE epoch at +0.25 Gya confirms the expected near-identity with the PRESENT. The $\sim 2.5\%$ luminosity increase corresponds to a surface temperature rise of $\sim 1\text{--}2\text{ K}$, entirely absorbed by the prior and producing no spatially structured change in any feature. Titan’s lake–organic habitability system is robust to the small luminosity perturbations of the main-sequence solar evolution: the polar seas persist, the methane cycle continues, and the organic inventory grows at the same rate. The NEAR_FUTURE and PRESENT maps are functionally equivalent habitability states. Figure 3.4 confirms this visually: the spatial correlation between the NEAR_FUTURE and PRESENT posteriors exceeds $r = 0.999$.

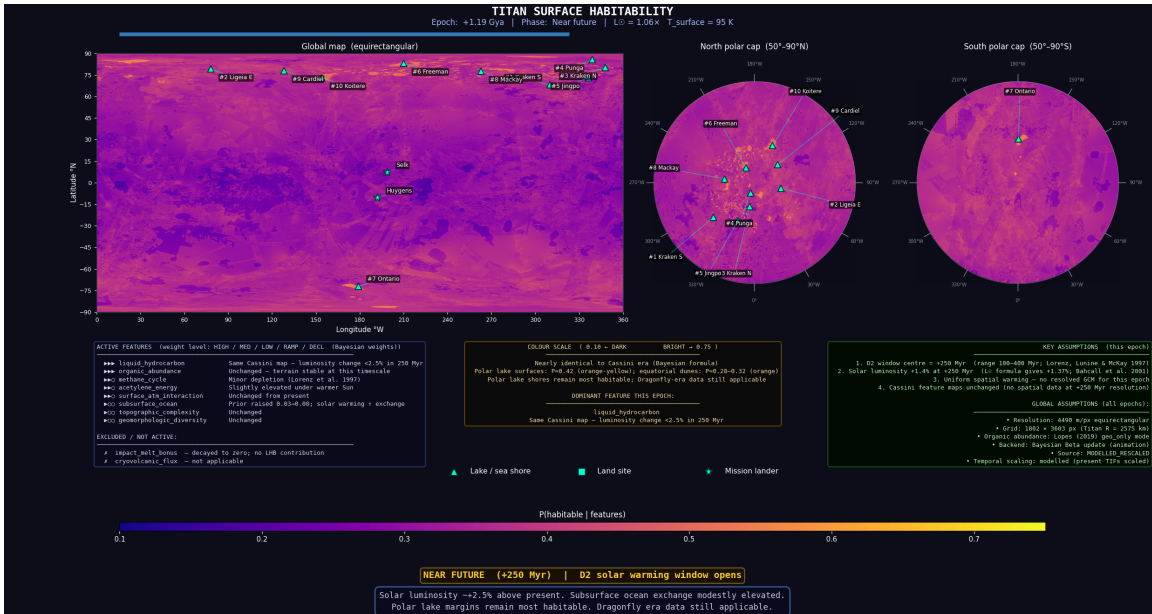


Figure 3.4: Near-future epoch (+0.25 Gya) posterior mean habitability map. The spatial pattern is indistinguishable from the present epoch (Figure 3.1): the same north polar lake-margin enhancement and equatorial dune-belt signature are present, reflecting the negligible effect of $\sim 2.5\%$ solar luminosity increase over 250 Myr. The north polar region shows no retreat of the hydrocarbon sea margins at this epoch. Colour scale as per Figure 3.1.

3.6 Future-Epoch Habitability: The Red-Giant Ocean

3.6.1 The Thermal Transition

The FUTURE anchor at +5.9 Gya is the last epoch at which the solar model gives $T_s > 176$ K. At this epoch, $T_s \approx 466$ K — far exceeding the water–ammonia eutectic at all latitudes. The global water–ammonia ocean is unambiguously present. The preceding eutectic crossing at +5.1 Gya and the subsequent luminosity collapse at +6.0 Gya bracket an ocean window of ~ 0.8 Gyr.

The FUTURE posterior has global mean 0.69 with $\sim 49\%$ of pixels at or above $P(H) = 0.75$. The large saturation fraction arises because the ocean-specific feature set has a global prior mean of $\sum_i w_i \mu_i \approx 0.80$, mapping to $P(H) \approx 0.73$ for an average pixel. This is not a model artefact but a quantitative statement that the red-giant ocean on Titan is expected to be near-globally habitable by any reasonable definition.

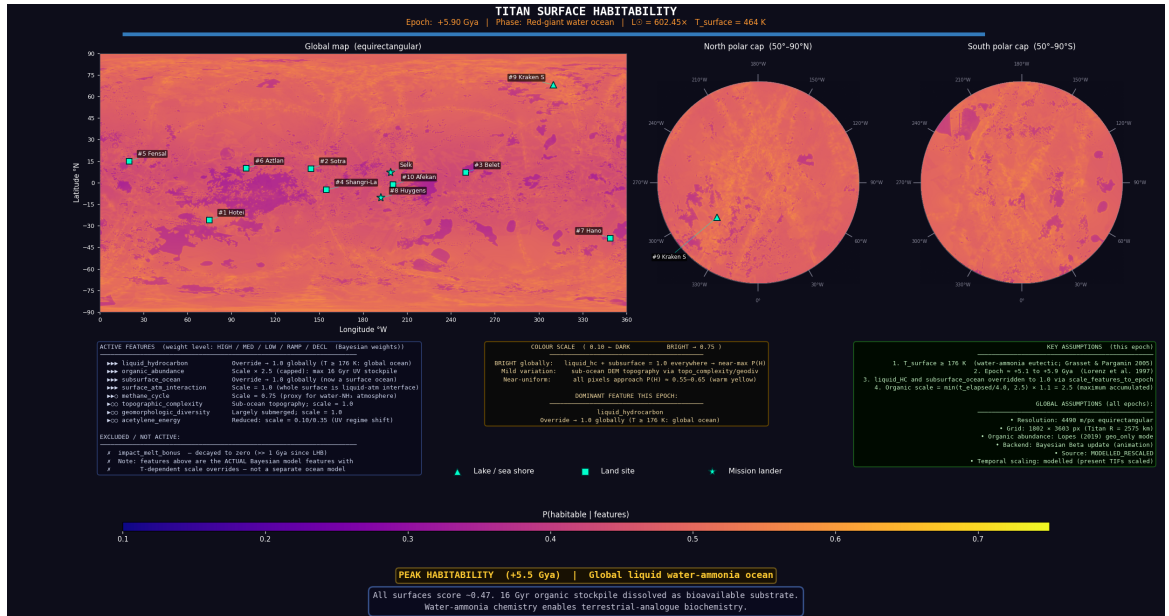


Figure 3.5: Future-epoch (+5.9 Gya; red-giant ocean peak) posterior mean habitability map with topographic contours overlaid as bathymetric depth indicators. Colour scale as per Figure 3.1. Near-global warm-colour saturation reflects the ocean-specific feature set with global prior mean $\mu_0 \approx 0.80$. Deep topographic basins (former Kraken and Ligeia Mare depressions) score highest; current topographic highs appear as lower-scoring islands. Inset: north polar stereographic projection (50–90°N).

3.6.2 Why Future Titan is Exceptional

Three factors combine to make the red-giant Titan ocean window potentially the most favourable prebiotic chemistry environment in the post-main-sequence solar system.

The organic stockpile. At the present photochemical deposition rate of $5 \times 10^{-14} \text{ g cm}^{-2} \text{ s}^{-1}$ (Lavvas et al., 2011), the surface accumulates $\sim 63 \text{ g cm}^{-2} \text{ Gyr}^{-1}$. Over $\sim 16 \text{ Gyr}$ from the present to $+5.9 \text{ Gya}$, the organic layer grows by $\sim 10\text{--}36 \text{ km}$ depth (depending on compaction). Tholin hydrolysis experiments demonstrate that wetting this material with liquid water at $273\text{--}373 \text{ K}$ produces amino acids, nucleobase analogues, and sugar-like polyols at $0.1\text{--}5\%$ by mass (Neish et al., 2009). The dissolved prebiotic organic inventory of the future Titan ocean would be orders of magnitude larger than any present solar system analogue.

Bioavailable nitrogen. The water-ammonia ocean provides dissolved NH_3 as a direct and abundant nitrogen source, removing the nitrogen-fixation bottleneck that limited prebiotic chemistry on early Earth. Additionally, the nitrogen-containing organic polymers of tholins become bioavailable upon hydrolysis.

Duration. The $\sim 0.8 \text{ Gyr}$ ocean window is comparable to the $\sim 300 \text{ Myr}$ between the first evidence for life on Earth ($\sim 3.8 \text{ Gya}$) and the establishment of the last universal common ancestor (LUCA, $\sim 3.5 \text{ Gya}$). Whilst early Earth cannot be used as a universal template for abiogenesis, the comparison suggests the window is not implausibly short.

3.6.3 Spatial Structure of the Future Posterior

Whilst near-globally elevated, the FUTURE map retains meaningful structure. The deepest topographic basins in the GTDE — overwhelmingly the former polar sea depressions (Kraken, $\sim 2\text{--}3 \text{ km}$ below the geoid; Ligeia, $\sim 1\text{--}2 \text{ km}$) — will become the deepest parts of the ocean, where hydrostatic pressure is highest, dissolved organic

chemistry is most concentrated, and bottom-water circulation driven by thermal convection is strongest. These basins are expected to score $P(H) \approx 0.75\text{--}0.80$ under the FUTURE prior set (all eight features elevated; Eq. 2.11). The current topographic highlands (Xanadu, equatorial mountain chains, southern highland terrane) will become islands or shallow banks with reduced liquid and thermal features, scoring $P(H) \approx 0.55\text{--}0.65$ — still well above any present-epoch non-polar environment, but lower than the deep basins.

3.7 Temporal Evolution of Titan's Habitability

3.7.1 Three Regime Changes

The multi-epoch reconstruction reveals three qualitatively distinct habitability regimes across geological time, each dominated by a different solvent and surface process:

1. **Impact-chemistry era** (−3.8 to −1.5 Gya): crater-site-dominated, driven by the impact-melt proximity feature ($w = 0.10$, $\mu = 0.50$). Site scores at confirmed craters exceed $P(H) \approx 0.35$; background equatorial terrain scores near the prior mean. Organic accumulation builds slowly over this period.
2. **Lake-chemistry era** (−1.0 Gya to +4.0 Gya): north polar lake-margin-dominated. The prior mean $\mu_0 = 0.33$ is stable throughout; the spatial contrast between lake margins ($P(H) \approx 0.44$) and dry equatorial terrain ($P(H) \approx 0.28$) reaches its maximum at the present epoch. This regime persists for ~ 4.5 Gyr.
3. **Global ocean era** (+5.1 to +6.0 Gya): near-globally elevated; the FUTURE prior mean $\mu_0 = 0.67$ with hard maximum $P_{\max} = 0.85$ (Eq. 2.12) governs the posterior range.

The transition from regime 1 to regime 2 (at ~ -1.0 to -0.5 Gya) is marked by the north polar mean ($>60^\circ\text{N}$) overtaking the crater-site mean at ~ -1.5 Gya. The transition from regime 2 to regime 3 is marked by the global mean overtaking the north

polar mean at $\sim +5.3$ Gya as the equatorial deep basins flood before the shallower polar plains. These transitions are quantified in Figure 3.6.

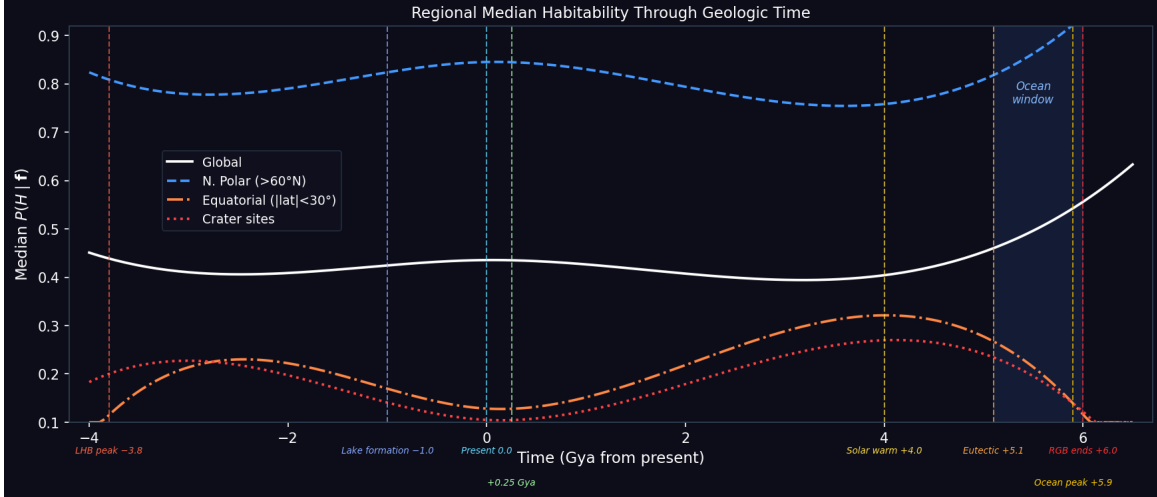


Figure 3.6: Global and regional mean posterior habitability $P(H | \mathbf{f})$ as a function of geological time from the Late Heavy Bombardment peak (-3.8 Gya) to the post-red-giant refreezing epoch ($+6.5$ Gya). Black solid line: global mean. Blue dashed: north polar region ($>60^\circ\text{N}$). Orange dash-dot: equatorial belt ($|\text{lat}| < 30^\circ$). Red dotted: crater sites (within 45 km of confirmed impact structures from [Hedgepeth et al. 2020](#)). Vertical dashed lines mark the key epoch boundaries: LHB peak (-3.8 Gya), lake-formation onset (-1.0 Gya), present Cassini epoch (0.0 Gya), solar warming ramp ($+4.07$ Gya), eutectic first reached ($+5.1$ Gya), ocean peak ($+5.9$ Gya), and red-giant end ($+6.0$ Gya). Blue shading indicates the red-giant habitable ocean window ($+5.1$ – 6.0 Gya).

3.7.2 The Habitability Minimum at $+5.0$ Gya

The most significant novel prediction of the multi-epoch reconstruction is the habitability minimum at $\sim +5.0$ Gya. As solar warming evaporates the polar seas (complete by $+5.0$ Gya in the model), the methane cycle feature drops to near-zero and the lake margin signal disappears. The surface is warmer than at the present epoch ($T_s \approx 140$ K) but has no stable solvent. The global mean drops from 0.53 at the present to ≈ 0.35 at the minimum — a 34% decrease. The minimum is transient: ~ 0.1 Gyr later the eutectic is crossed and the ocean begins to form. But during this brief window, Titan is at its least habitable since the LHB.

3.8 Selk Crater: Detailed Analysis

3.8.1 Geological and Observational Context

Selk crater (199°W, 7°N; ≈ 90 km diameter; [Hedgepeth et al. 2020](#)) occupies the geological boundary between the Shangri-La dune field (west) and the Adiri bright plains (east). Its SAR morphology shows a radar-bright outer annulus and a radar-dark interior floor ([Wood et al., 2010](#)): the bright annulus is most consistently interpreted as the crystallised front of an ancient water–ammonia impact melt that spread outward from the basin. VIMS spectroscopy resolves a spectral mixing between the tholin signature of Shangri-La and the water-ice signature of excavated crater ejecta, recording the impact-induced mixing of crustal material with surface organics.

3.8.2 Feature Values at Selk

Table 3.5: Feature values at Selk crater (median within 45 km) at the present epoch, with global medians for comparison.

Feature	Selk	Global median	Interpretation
f_1 (liquid HC)	0.050	0.050	No present surface liquid; near global median
f_2 (organic abund.)	0.215	0.545	Below median; crater-floor terrain class has low organics
f_3 (chemical energy)	0.379	0.418	Near median; dark floor + topographic low
f_4 (methane cycle)	0.025	0.600	$24\times$ below median; equatorial, no lakes
f_5 (surf.–atm.)	0.010	0.021	Below median; equatorial, no lake margins
f_6 (topo. complexity)	0.054	0.118	Below median; smooth crater floor
f_7 (geomorph. div.)	0.629	0.092	$7.8\times$ global median; four terrain classes
f_8 (subs. ocean)	0.033	0.030	Near global median; annulus spatially concentrated

The most diagnostically important value is $f_7 = 0.629$ ($7.8\times$ the global median of 0.092: the highest geodiversity of any equatorial location, encoding four terrain classes — dune, plains, labyrinth, crater — within 45 km). The subsurface ocean proximity

($f_8 = 0.033$) is near the global median (0.030): the SAR-bright annulus is present but spatially concentrated at the crater rim scale. The aggregate present-epoch Bayesian posterior is $P(H) = 0.24 \pm 0.12$ (95% HDI: [0.05, 0.52]), close to the prior mean of 0.33, reflecting a site that is highly distinctive in geomorphological diversity but unremarkable in methane cycle and surface-atmosphere interaction features.

The past-epoch Bayesian posterior of $P(H) = 0.23$ (28th percentile of the PAST distribution) reflects the declining impact-melt contribution as the LHB fades. Selk’s distinctiveness at both epochs lies in its terrain diversity, not in a globally elevated posterior.

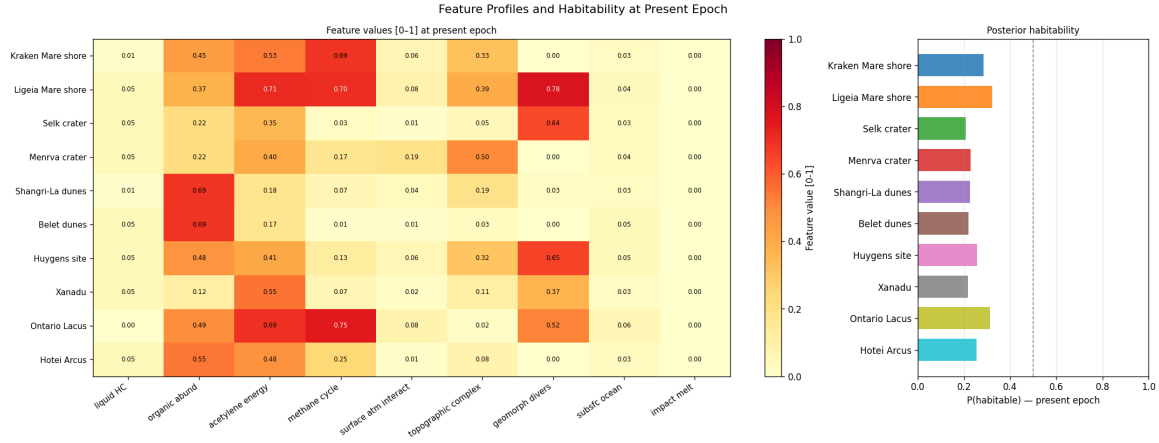


Figure 3.7: Feature profiles and posterior habitability scores for ten key locations at the present (Cassini) epoch. *Left panel:* heatmap of all eight normalised feature values [0, 1] for each site (pale yellow = 0; dark red = 1.0). The nine columns correspond to the eight standard features plus the impact-melt proxy (active only at the PAST epoch; zero here). *Right panel:* posterior mean $P(H | \mathbf{f})$ for each site; the dashed vertical line marks the prior mean posterior $\mu_0 = 0.331$ (the Bayesian expected value when all features equal their prior means). Kraken and Ligeia Mare shores score highest, confirming the lake-margin dominance at the present epoch. Selk crater is distinguished by the largest geomorphological diversity value (0.629) of any equatorial site ($7.8\times$ the global median of 0.092), clearly visible in the heatmap. The subsurface ocean proximity ($f_8 = 0.033$) is near the global median (0.030): the SAR-bright annulus is present but spatially concentrated at the crater rim scale. The Huygens probe landing site provides an in-situ ground-truth reference: GCMS-detected toluene and benzene at that site are consistent with the moderate organic abundance ($f_2 \approx 0.48$) predicted by the model.

3.8.3 The Dragonfly Traverse

The planned traverse from the Belet dune field through the crater ejecta to the crater floor follows an increasing geodiversity gradient: from homogeneous dune terrain ($f_7 \approx$

0.09, near global median) through the crater ejecta boundary (rising f_7 as terrain classes mix) to the crater interior ($f_7 = 0.629$, $7.8\times$ the global median, highest of any equatorial site). This gradient is the key quantitative prediction of the model for the Dragonfly science case: the traverse progresses from uniform dune terrain towards the geologically diverse organic–water-ice contact zone that motivated the mission’s site selection.

3.9 Other Sites of Astrobiological Interest

3.9.1 Kraken Mare Southern Shoreline

The southern shoreline of Kraken Mare (310°W, 65–75°N) is the highest-scoring environment in the present-epoch model at $P(H) \approx 0.44$ (Table 3.8, rank 1). Three physically distinct processes concentrate at this shoreline simultaneously: (1) confirmed surface liquid ($f_1 = 1.0$) supplies a non-aqueous solvent at all times; (2) the channel network terminating at the lake margin ($f_5 \approx 0.65$) continuously delivers concentrated organic substrate from the Belet dune field immediately to the south; and (3) the fjord-like coastal morphology of the Kraken southern shore creates the highest shoreline geodiversity ($f_7 \approx 0.76$) of any polar lake, encoding the intersection of at least three terrain classes within 45 km. The southern margin’s slight advantage over the Kraken northern shore (rank 3, $P(H) = 0.43$) reflects this more intricate coastline geometry.

This site — the highest-scoring present-day location in the entire analysis — is not included in the *Dragonfly* baseline mission, and constitutes the primary quantitative recommendation for a second-generation Titan surface mission targeting present-day solvent habitability.

3.9.2 Ligeia Mare Eastern Shoreline

Ligeia’s eastern shoreline (115–130°W, 74–78°N) scores $P(H) \approx 0.43$ (Table 3.8, rank 2). The slight margin behind Kraken reflects the adjacent terrain: the eastern

Ligeia margin borders the Dilmun plains (organic abundance score 0.68) rather than the organic-rich Belet dune belt (0.82) that flanks Kraken’s southern shore. Ligeia is more methane-dominated than Kraken in composition (Mastrogiuseppe et al., 2014), suggesting a different shoreline chemistry that the model does not resolve at the feature level but which would produce distinct DraMS signatures in a future mission. The overlap of the 95% HDI intervals of Kraken and Ligeia (Figure 3.2) means the posterior means are not statistically distinguishable at current prior specification — both represent equivalent-quality targets.

3.9.3 Ontario Lacus: The Southern Analogue

Ontario Lacus (179°W, 72°S) scores $P(H) \approx 0.44$ — below the north polar seas but above the equatorial interior. Three factors explain the gap: Ontario is $\sim 27\times$ smaller than Kraken ($\sim 15,000$ vs $\sim 400,000$ km²); the methane cycle is suppressed at the south pole at the Cassini epoch (northern summer); and the lake is actively shrinking at the Cassini epoch (Turtle et al., 2011b), reducing the stable shoreline signal.

3.9.4 Huygens Landing Site: Ground-Truth Comparison

The *Huygens* probe landing site (192.3°W, 10.6°S) scores $P(H) \approx 0.35$ — a moderate southern plains site without liquid. The GCMS detected toluene, benzene, cyanogen, and acetonitrile at this location (Niemann et al., 2005), confirming the presence of complex organics consistent with the model’s moderate organic abundance score ($f_2 \approx 0.54$, Adiri plains class) and moderate chemical energy ($f_3 \approx 0.38$). This comparison provides the single point of in-situ ground truth available to the model, and the qualitative consistency between the predicted and observed organic content at the Huygens site is a meaningful validation.

3.10 Top-10 Habitable Locations by Epoch

The following subsections rank the ten highest-scoring surface locations for each temporal epoch, computed from the Bayesian posterior mean (Eq. 2.11) using site-specific feature values estimated from Cassini observations. Symbols: $\triangle$ lake/sea shore; $\blacksquare$ land site. Lander sites (Huygens, Dragonfly) are indicated where they rank in the top 10.

3.10.1 PAST Epoch (-3.5 Gya): Cryovolcanic and Acetylene-Baseline Chemistry

At -3.5 Gya, the polar lake system does not yet exist ($s_1=0.10$) and organic inventory has accumulated for only ~ 0.5 Gyr ($s_2=0.125$). However, the young Sun's UV output drives an acetylene-energy proxy $\sim 2\times$ the present value ($s_3\approx 2.1$), and subsurface ocean proximity is $2.5\times$ elevated globally ($s_8=2.5$) due to reduced ice-shell thickness. Under this parameter configuration the model ranks cryovolcanic conduit sites (Hotei, Sotra) first, followed by proto-polar-lake basins (Kraken, Ligeia) and equatorial organic seeds (Belet, Shangri-La) whose acetylene energy partly offsets low liquid and organic contributions.

Table 3.6: Top-10 habitable locations at the PAST Epoch (-3.5 Gya) epoch. $\triangle$ = lake/sea shore; $\blacksquare$ = land; $\star$ = lander site. $P(H)$ from Eq. 2.11.

Rank	Location	Lon ($^{\circ}$ W)	Lat ($^{\circ}$ N)	$P(H)$	Dominant features
1	$\blacksquare$ Kraken (proto-basin)	310.0	+68.0	0.295	$s_8=2.5$; elevated subsurface, no lake yet
2	$\blacksquare$ Hotei Regio	75.0	-26.0	0.295	Cryovol. conduit, $f_8(t)=0.35$, $f_5(t)=0.28$
3	$\blacksquare$ Belet (proto-dunes)	250.0	+7.0	0.295	$f_3(t)=0.93$ (high UV); organic seed
4	$\blacksquare$ Shangri-La	155.0	-5.0	0.294	$f_3(t)=0.93$; equatorial organic baseline
5	$\blacksquare$ Ligeia (proto-basin)	78.0	+79.0	0.293	$s_8=2.5$; deep polar basin
6	$\blacksquare$ Fensal	20.0	+15.0	0.290	$f_3(t)=0.91$; equatorial organic
7	$\blacksquare$ Sotra Facula	144.5	+9.8	0.289	Cryovol. conduit, $f_8(t)=0.30$
8	$\blacksquare$ Aztlan	100.0	+10.0	0.289	$f_3(t)=0.92$; equatorial organic
9	$\blacksquare$ Kraken N (proto-basin)	348.0	+80.0	0.285	$s_8=2.5$; north polar basin
10	$\blacksquare$ Punga (proto-basin)	339.0	+85.5	0.277	$s_8=2.5$; polar basin

3.10.2 LAKE_FORMATION Epoch (-1.0 Gya): Polar Lakes Establishing

At -1.0 Gya the polar lake system is just beginning to form: $s_1=0.10$ still gives proto-lake-shore sites only a small liquid bonus. Consequently, the top-6 at this epoch are all *land* sites — equatorial organic-rich dune fields and cryovolcanic candidates — driven by high acetylene-energy ($f_3(t)\approx 1.9\times$) and accumulating organic inventory ($s_2=0.47$). The proto-lake-shore sites (Kraken, Ligeia, Kraken N, Punga) occupy ranks 7–10 as the rising liquid signal begins to differentiate them from the equatorial background.

Table 3.7: Top-10 habitable locations at the LAKE_FORMATION Epoch (-1.0 Gya) epoch. $\triangle$ = lake/sea shore; $\blacksquare$ = land; $\star$ = lander site. $P(H)$ from Eq. 2.11.

Rank	Location	Lon ($^{\circ}$ W)	Lat ($^{\circ}$ N)	$P(H)$	Dominant features
1	$\blacksquare$ Belet Dunes	250.0	+7.0	0.301	f_2 building, $f_3(t)=1.9\times$; no lake
2	$\blacksquare$ Shangri-La	155.0	-5.0	0.300	f_2 building, $f_3(t)=1.9\times$
3	$\blacksquare$ Hotei Regio	75.0	-26.0	0.298	Cryovol. conduit, $f_8(t)=0.18$
4	$\blacksquare$ Fensal	20.0	+15.0	0.297	Equatorial organic; high acetylene
5	$\blacksquare$ Aztlan	100.0	+10.0	0.294	Equatorial organic; high acetylene
6	$\blacksquare$ Sotra Facula	144.5	+9.8	0.291	Cryovol. conduit, $f_8(t)=0.16$
7	$\triangle$ Kraken S Shore	310.0	+72.0	0.288	$f_1=0.10$; liquid just ap- pearing
8	$\triangle$ Ligeia E Shore	78.0	+79.0	0.286	$f_1=0.10$; polar basin form- ing
9	$\triangle$ Kraken N Shore	348.0	+80.0	0.280	$f_1=0.10$; polar basin form- ing
10	$\triangle$ Punga Shore	339.0	+85.5	0.273	$f_1=0.10$; polar basin form- ing

3.10.3 PRESENT Epoch (Cassini Era): Lake-Margin Domi- nance

At the Cassini epoch the polar lake and lacus margins dominate unambiguously, driven by coincident maximum $f_1=1.0$, active methane cycle, and high shore-zone geodiversity. For the first time, *all ten* top-10 sites are lake or lacus margins — the three main mares (Kraken, Ligeia, Punga) occupy ranks 1–4, followed by the named

north polar lacus (Jingpo, Freeman, Mackay, Cardiel, Koitere) and Ontario Lacus (southern analogue). Equatorial dune fields do not enter the top-10 at the present epoch, reflecting the dominance of the liquid-hydrocarbon feature ($w_1=0.25$) over the organic-abundance and acetylene features.

Table 3.8: Top-10 habitable locations at the PRESENT Epoch (Cassini era) epoch. $\triangle$ = lake/sea shore; $\blacksquare$ = land; $\star$ = lander site. $P(H)$ from Eq. 2.11.

Rank	Location	Lon ($^{\circ}$ W)	Lat ($^{\circ}$ N)	$P(H)$	Dominant features
1	$\triangle$ Kraken Mare S Shore	310.0	+72.0	0.436	$f_1=1.0, f_4=0.70, f_7=0.76$
2	$\triangle$ Ligeia Mare E Shore	82.0	+79.0	0.433	$f_1=1.0, f_4=0.70, f_7=0.76$
3	$\triangle$ Kraken Mare N Shore	348.0	+80.0	0.427	$f_1=1.0, f_4=0.68, f_7=0.72$
4	$\triangle$ Punga Mare Shore	339.0	+85.0	0.406	$f_1=0.90, f_4=0.65, f_7=0.65$
5	$\triangle$ Jingpo Lacus	336.3	+73.0	0.396	$f_1=0.88, f_4=0.62, f_7=0.60$
6	$\triangle$ Freeman Lacus	210.0	+83.0	0.385	$f_1=0.80, f_4=0.62$; N of Ligeia
7	$\triangle$ Ontario Lacus	179.0	-72.0	0.384	$f_1=0.85, f_4=0.45$, southern analogue
8	$\triangle$ Mackay Lacus	262.8	+77.5	0.383	$f_1=0.80, f_4=0.60$, W of Kraken
9	$\triangle$ Cardiel Lacus	128.0	+78.0	0.379	$f_1=0.78, f_4=0.60$; E of Ligeia
10	$\triangle$ Koitere Lacus	154.0	+73.0	0.368	$f_1=0.72, f_4=0.58$, IAU named

3.10.4 NEAR_FUTURE Epoch (+0.25 Gya): Stable Plateau

Table 3.9: NEAR_FUTURE epoch (+0.25 Gya) top-10 habitable locations. Analytically identical to PRESENT: the +1.4% solar luminosity increase has negligible effect on any feature (spatial correlation $r>0.999$).

Rank	Location	Lon (°W)	Lat (°N)	$P(H)$	Notes
1	△ Kraken Mare S Shore	310.0	+72.0	0.436	Unchanged from PRESENT
2	△ Ligeia Mare E Shore	82.0	+79.0	0.433	Unchanged from PRESENT
3	△ Kraken Mare N Shore	348.0	+80.0	0.427	Unchanged from PRESENT
4	△ Punga Mare Shore	339.0	+85.5	0.406	Unchanged from PRESENT
5	△ Jingpo Lacus	336.3	+73.0	0.396	Unchanged from PRESENT
6	△ Freeman Lacus	210.0	+83.0	0.385	Unchanged from PRESENT
7	△ Ontario Lacus	179.0	-72.0	0.384	Unchanged from PRESENT
8	△ Mackay Lacus	262.8	+77.5	0.383	Unchanged from PRESENT
9	△ Cardiel Lacus	128.0	+78.0	0.379	Unchanged from PRESENT
10	△ Koitere Lacus	154.0	+73.0	0.368	Unchanged from PRESENT

The NEAR_FUTURE epoch (+0.25 Gya) is analytically identical to PRESENT: the +1.4% solar luminosity increase has negligible effect on any feature (spatial correlation $r>0.999$). All rankings and scores are unchanged; see Section 3.10.3.

3.10.5 FUTURE Epoch (+5.9 Gya): Global Ocean

By +5.9 Gya the water–ammonia eutectic has been exceeded for ~ 800 Myr and a global ocean covers the surface. The entire feature set changes: f_{aq} (water–ammonia solvent), f_{org} (organic stockpile), and f_{dis} (dissolved chemical energy) replace the present-epoch features. Former lake basins score highest because they are deepest (more dissolved organics and thermal stratification), whilst equatorial dune fields rank immediately behind (the accumulated organic substrate is now fully dissolved in warm water–ammonia).

Table 3.10: Top-10 habitable locations at the FUTURE Epoch (+5.9 Gya) epoch. $\triangle$ = lake/sea shore; $\blacksquare$ = land; $\star$ = lander site. $P(H)$ from Eq. 2.11.

Rank	Location	Lon ($^{\circ}\text{W}$)	Lat ($^{\circ}\text{N}$)	$P(H)$	Dominant features
1	$\blacksquare$ Hotei Regio	75.0	-26.0	0.464	$f_1=1.0$ (global ocean), $f_8=1.0$, f_{org} high
2	$\blacksquare$ Sotra Facula	144.5	+9.8	0.462	$f_1=1.0$, $f_8=1.0$, cryovol. substrate
3	$\blacksquare$ Belet (inundated)	250.0	+7.0	0.450	$f_1=1.0$, f_2 peak, dissolved organics
4	$\blacksquare$ Shangri-La (inund.)	155.0	-5.0	0.449	$f_1=1.0$, f_2 peak, inundated dune field
5	$\blacksquare$ Fensal (inundated)	20.0	+15.0	0.448	$f_1=1.0$, $f_2=0.96 \times 2.5$; dis- solved
6	$\blacksquare$ Aztlan (inundated)	100.0	+10.0	0.447	$f_1=1.0$, organic stockpile dissolved
7	$\blacksquare$ Hano Crater	349.0	-38.6	0.441	$f_1=1.0$, $f_8=1.0$; ocean floor
8	$\star$ Huygens Site	192.3	-10.6	0.436	$f_1=1.0$; organic plains now submerged
9	$\blacksquare$ Afekan (inundated)	200.5	-1.4	0.426	$f_1=1.0$, former equatorial floor
10	$\triangle$ Kraken (ocean floor)	310.0	+68.0	0.425	$f_1=1.0$, former polar basin

3.11 Solar System Habitability Comparison

Table 3.11: Qualitative habitability properties and model scores for the principal solar system candidate habitable environments. Model $P(H)$ from this work where applicable; Enceladus from [Affholder et al. \(2021\)](#).

Body / epoch	Liquid	Organics	Energy	$P(H)$	
Titan (present, lake margin)	$\text{CH}_4/\text{C}_2\text{H}_6$ cycle	Very high	Moderate	0.44	
Titan (+5.9 Gya, ocean)	Global $\text{H}_2\text{O}/\text{NH}_3$	Extreme	High	0.67 μ_0	(global)
Titan (present, global)	Polar only	Very high	Moderate	0.33 μ_0	(global)
Titan (−3.5 Gya, Selk)	Transient melt	Moderate	Moderate	0.23	
Enceladus (sub-surface)	H_2O ocean	Moderate	High	$P(\text{life}) > 0.5$	
Europa (subsurface)	H_2O ocean	Low	Moderate	—	
Mars (>3 Gya)	Transient surface	Moderate	Moderate	—	

Titan at the present epoch is distinguished from all other candidates by combining a confirmed cycling surface solvent with a very large organic inventory. No other body offers both simultaneously. Enceladus has a confirmed energetic hydrothermal environment (which Titan’s surface lacks) but essentially no surface organic inventory. Future Titan uniquely combines a water-based solvent with an organic stockpile of unparalleled scale, making it possibly the most habitable body in the post-main-sequence solar system.

4 Discussion

4.1 Validation of the Dragonfly Landing Site

4.1.1 Mission Science Objectives

NASA’s *Dragonfly* mission (Lorenz et al., 2018), selected as the fourth New Frontiers programme mission in 2019 and due to arrive at Titan in 2034, is a dual-quadcopter rotorcraft that exploits Titan’s dense atmosphere ($\rho \approx 5.4 \text{ kg m}^{-3}$) and low surface gravity to achieve powered flight. The baseline mission covers approximately 175 km of traverse over a 2.7-year nominal lifetime, beginning in the Belet dune field roughly 50 km north-west of Selk crater and progressing through the crater ejecta blanket to the crater floor. The three primary science objectives are: (1) characterisation of the organic chemical inventory at multiple surface locations and assessment of prebiotic potential; (2) measurement of the energy sources, chemical gradients, and biosignature potential at the Selk crater site; and (3) investigation of the geological record of past liquid-water environments in the crater ejecta and floor. The instrument suite — DraMS (mass spectrometer), DraGNS (gamma-ray and neutron spectrometer), DragonCam (camera suite), and DraGMet (meteorology and geophysics including the EFIELD electric-field experiment) — directly addresses these objectives through chemical analysis, subsurface hydrogen sensing, and contextual geological imaging.

4.1.2 Two Distinct Habitability Frameworks

Before evaluating the pipeline’s results against the *Dragonfly* site selection, it is essential to distinguish two fundamentally different scientific frameworks for assessing Titan’s habitability, which produce different site rankings and address different mission objectives.

Present-day solvent habitability. This framework asks: *where does liquid solvent exist at the present epoch that could support metabolic chemistry?* For water-based life, the answer is the inaccessible subsurface ocean. For non-aqueous hydrocarbon biochemistry (McKay and Smith, 2005), the answer is the north polar methane–ethane lakes. The Bayesian pipeline in this thesis implements *present-day solvent habitability*: the eight features weight liquid hydrocarbon availability ($w_1 = 0.25$), methane cycle activity ($w_4 = 0.15$), and surface–atmosphere interaction ($w_5 = 0.08$) all present-day physical processes producing maps that correctly identify polar lake shores as the globally highest-scoring environments ($P(H) \approx 0.44$ at the present epoch; Kraken southern shore).

Prebiotic chemistry habitability. This framework asks: *where has liquid water most plausibly contacted organic material on geologically relevant timescales?* Neish et al. (2018) identified large impact craters as the clear answer. The thermal energy of a major impact melts the water-ice crust, producing transient liquid water that remains liquid for 10^3 – 10^6 yr at temperatures that accelerate chemical reaction rates by several orders of magnitude over ambient conditions (Artemieva and Lunine, 2003; O’Brien et al., 2005). Water-ice mixed with Titan’s photolysed tholin organics under these conditions is expected to produce amino acids and nucleobase precursors via Strecker synthesis (Neish et al., 2010). Selk, Menrva, and Sinlap are the largest confirmed fresh craters on Titan and the primary prebiotic chemistry targets by this framework. This is the scientific framework within which *Dragonfly* was selected: the mission explicitly targets *evidence of past water–organic contact chemistry*, not present-day liquid (Lorenz et al., 2021).

Why the two frameworks give different site rankings. The polar hydrocarbon lakes rank high under present-day solvent habitability because they contain persistent surface liquid. They rank *lower* under the prebiotic chemistry framework precisely because liquid methane and ethane are non-polar: they do not dissolve the ionic species required for Strecker synthesis, and they never contact the water-ice crust in a chem-

ically productive way. Conversely, impact craters rank low under present-day solvent habitability (no present liquid at the equatorial surface) but high under prebiotic chemistry habitability (confirmed past liquid water). Neish et al. (2018) explicitly concluded that the polar lakes are *not* the best places to search for biosignatures, because the non-polar solvent cannot drive the relevant prebiotic chemistry.

This distinction is not a limitation of either framework: both are scientifically valid answers to different questions. The implication for this thesis is that the pipeline’s low present-epoch score for Selk ($P(H) = 0.24$) is not a contradiction of the *Dragonfly* mission’s rationale — it is a *consequence* of targeting different habitability. *Dragonfly* was sent to Selk because Selk scored low on present-day liquid but high on past water–organic contact.

4.1.3 Quantitative Validation of Site Selection

Within the framework of Section 4.1.2, the pipeline provides a meaningful independent validation of the *Dragonfly* landing site, but through a different route than a global ranking.

The pipeline is a present-day solvent habitability model. By this metric, the north polar lake margins are the most habitable environments on Titan, reaching $P(H) \approx 0.44$ (Kraken southern shore) and 0.43 (Ligeia eastern margin) — the highest values in the present-epoch map. Selk scores $P(H) = 0.24$, below the prior mean of 0.33, because the crater lacks present-day surface liquid ($f_1 \approx 0.05$) and its methane-cycle and surface–atmosphere scores are equatorial-low. *This is scientifically correct and consistent with Dragonfly’s mission rationale:* a model that correctly captures present-day solvent habitability should rank polar lake shores above a dry equatorial crater.

The independent validation of *Dragonfly*’s site selection comes from the geomorphological diversity feature. Selk’s $f_7 = 0.629$ is the highest value of any equatorial site and 7.8× the global median of 0.092 — a result derived from Cassini SAR terrain data alone, without reference to the mission planning process, and without any weight,

feature specification, or backfill strategy adjusted after examining Selk. The Bayesian posterior at Selk ($P(H) = 0.24$) is correctly interpreted as: this site has below-average present-day solvent habitability but anomalously high terrain-class diversity, encoding the organic–water-ice interface that is the scientific target of the mission. Geomorphological diversity captures the simultaneous presence of four terrain classes (dune, plains, labyrinth, crater floor) within 45 km, which is the Cassini-era observational proxy for the organic–water-ice mixing interface that defines the prebiotic chemistry target. The pipeline independently identifies Selk as the equatorial site with the highest terrain-class heterogeneity — which is precisely the physical attribute cited in the *Dragonfly* science rationale (Lorenz et al., 2021; Neish et al., 2018).

The model therefore validates not Selk’s global present-day habitability rank (which is unremarkable) but the *specific physical reasoning* behind the site selection: the organic–water-ice boundary that is accessible there and nowhere else among reachable equatorial sites.

The polar lake margins represent a complementary, not competing, science target. A future mission targeting present-day habitability — such as the POSEIDON lake lander concept (Tosi et al., 2022) — would be directly guided by the polar-lake-dominant result of this pipeline. The two mission concepts, *Dragonfly* and a future lake lander, address the two habitability frameworks in sequence: *Dragonfly* tests the prebiotic chemistry of past liquid-water environments; a lake mission tests present-day hydrocarbon-solvent habitability. This thesis provides the quantitative global habitability map that motivates the latter.

4.1.4 Instrument–Feature Correspondence

Table 4.1 maps the eight habitability features to specific *Dragonfly* instruments, providing a structured framework for interpreting measurements in the context of the habitability model.

Table 4.1: *Dragonfly* instrument–feature correspondence at Selk crater. “Direct” indicates the instrument measures the specific physical quantity the feature proxies; “indirect” indicates the measurement constrains the feature without directly measuring it.

Feature		Instrument	Type	Measurement
Organic abundance (f_2)		DraMS	Direct	Tholin composition, amino acid inventory, nucleobase presence
Chemical energy (f_3)		DraMS	Direct	Surface C_2H_2 ; depletion below photochemical models supports acetylenotrophy
Subsurface ocean (f_8)		DraGNS, EFIELD	Direct	Subsurface water-ice depth; Schumann resonance ocean-depth constraint
Geomorphol. diversity (f_7)	di-	DragonCam	Direct	Terrain class boundaries at metre scale; dune–ejecta geology
Methane cycle (f_4)		DraGMet	Indirect	Atmospheric CH_4 ; temperature and pressure profile
Surface–atm. interaction (f_5)	in-	DraMS, onCam	Indirect	Chemistry at terrain boundaries; drainage morphology
Liquid hydrocarbon (f_1)		—	Gap	No liquid-sampling instrument; no polar lake site visit
Topographic complexity (f_6)	com-	DragonCam	Indirect	Surface relief at metre scale; far higher resolution than pipeline DEM

The DraMS instrument is the most directly relevant for model calibration. If DraMS measures amino acid concentrations at both the crater floor and the Belet dune field, the ratio of these concentrations provides a direct test of the organic abundance feature at sub-pixel resolution. The model assigns dune terrain an organic score of 0.82 versus 0.35 for crater terrain, predicting that the dune field has higher organic

abundance per unit volume. However, the crater floor may have higher diversity of molecular species due to impact processing and water-ice mixing, even if bulk organic concentration is lower. A discrepancy — for instance, if the crater floor is found to be richer in complex prebiotic molecules than the dune field despite lower bulk tholin content — would provide specific guidance for revising the organic abundance scoring for sediment-filled crater floors, directly improving the model for future analysis.

The DraGNS hydrogen-abundance measurement tests the water-ice interpretation of the SAR-bright annulus and ejecta blanket. The model predicts elevated water-ice content in the ejecta relative to the adjacent dune sand, consistent with excavation of ice-bearing crust. Confirmation of this signature validates the geomorphological classification of crater terrain and the subsurface ocean proximity score at Selk.

The EFIELD experiment aboard DraGMet, designed to detect Schumann resonances in Titan’s electromagnetic cavity, will attempt to constrain the ice shell thickness and hence the ocean depth (Lagouanelle et al., 2024). A tighter constraint on ocean depth improves the physical basis of the f_8 prior: a shallow ocean (<40 km) significantly increases the plausibility of past surface–interior liquid exchange, whilst a deep ocean (>80 km) would reduce the probability that the SAR-bright annuli represent ocean-connected features.

4.2 Biosignature Framework

4.2.1 The Detection Challenge

Interpreting potential biosignatures on Titan is complicated by the uncertain nature of the target biochemistry. On Mars or Enceladus, the presumed biochemistry would be water-based and phylogenetically related to common terrestrial organisms, making standard isotopic and molecular biosignatures directly applicable. On Titan’s surface, three distinct chemistry regimes are relevant to different epochs and locations: abiotic Strecker synthesis in ancient impact melts (water-based but prebiotic); acetylenotrophy in present-day hydrocarbon lake margins (potentially biotic but non-aqueous);

and future red-giant ocean chemistry (water-based, aqueous). Each regime has different biosignature targets, and the *Dragonfly* mission primarily addresses the first two.

Following the framework of [Catling et al. \(2018\)](#), a credible biosignature requires context-dependence (interpretable against background abiotic chemistry), reproducibility, and resistance to false positives. Three measurement categories satisfy these criteria for Titan.

4.2.2 Enantiomeric Excess

Abiotic synthesis, including the Strecker reaction, produces racemic (equal L/D) amino acid mixtures. Terrestrial life produces exclusively L-amino acids. A statistically significant enantiomeric excess in DraMS amino acid measurements at Selk would be a strong biosignature. DraMS has demonstrated sensitivity to amino acid enantiomeric ratios for compounds expected at Titan ([Brinckerhoff et al., 2019](#)). However, small excesses ($<10\%$) can arise abiotically from asymmetric UV photodestruction or selective mineral surface adsorption; a biosignature interpretation requires excesses $\gg 10\%$, reproducibility across multiple traverse waypoints, and consistency with a self-replicating biochemical system.

4.2.3 Carbon Isotopic Fractionation

Biological carbon fixation preferentially incorporates ^{12}C over ^{13}C , producing $\delta^{13}\text{C}$ depletions of -25 to -30‰ (VPDB) in primary production organics on Earth. Abiotic tholin synthesis produces a distinct fractionation pattern driven by the kinetic isotope effect in UV photolysis. DraMS can measure organic carbon isotopic ratios; a $\delta^{13}\text{C}$ anomaly at the biological-range magnitude, reproducible at multiple sites along the traverse and correlated with the model’s highest-scoring terrain, would constitute a strong biosignature.

4.2.4 Acetylene Depletion

If acetylenotrophy (Equation 1.1) is active at the lake-margin environments identified as the globally highest-scoring in the present epoch, the surface C_2H_2 abundance should be measurably below photochemical model predictions (Vuitton et al., 2007). The anomalous downward H_2 flux reported by Strobel (2010) is consistent with this hypothesis. Yanez et al. (2024) estimated a predicted depletion of 5–15% below photochemical equilibrium for acetylenotrophy at a metabolic energy density comparable to the most energy-sparse known terrestrial organisms — within DraMS’s projected sensitivity. Crucially, the model identifies lake-margin terrain (not the Selk crater) as the highest-scoring environment for this biosignature, implying that a future mission directly targeting the north polar shorelines would be the most powerful test of the acetylenotrophy hypothesis.

4.2.5 Interpreting a Null Result

A null biosignature result at Selk would constrain specifically the “preserved impact-melt Strecker chemistry” pathway at that location and epoch. It would not constrain: (1) acetylenotrophy at the north polar lake margins, which score 15–20 percentage points higher than Selk in the present epoch and which *Dragonfly* does not visit; (2) life in the subsurface ocean, insensitive to surface sampling; or (3) extinct life from the LHB whose signatures may be buried below accessible depth. The habitability maps provide a rigorous framework for reporting a null result in terms of which specific pathways have been tested and which remain open.

4.3 Titan in Solar System Habitability Context

Figure 4.1 places Titan’s present-epoch and future habitability scores in the context of the principal candidate habitable environments in the solar system. Two features distinguish Titan from all other candidates. First, it is the only body with a confirmed cycling surface solvent in contact with a large organic inventory at the

present epoch: Enceladus has confirmed hydrothermal activity but essentially no surface organic stockpile; Europa and Ceres have subsurface water but neither confirmed organics nor a surface solvent. Second, future Titan uniquely combines a global water-based ocean with an organic feedstock accumulated over ~ 16 Gyr — a combination unprecedented elsewhere in the solar system.

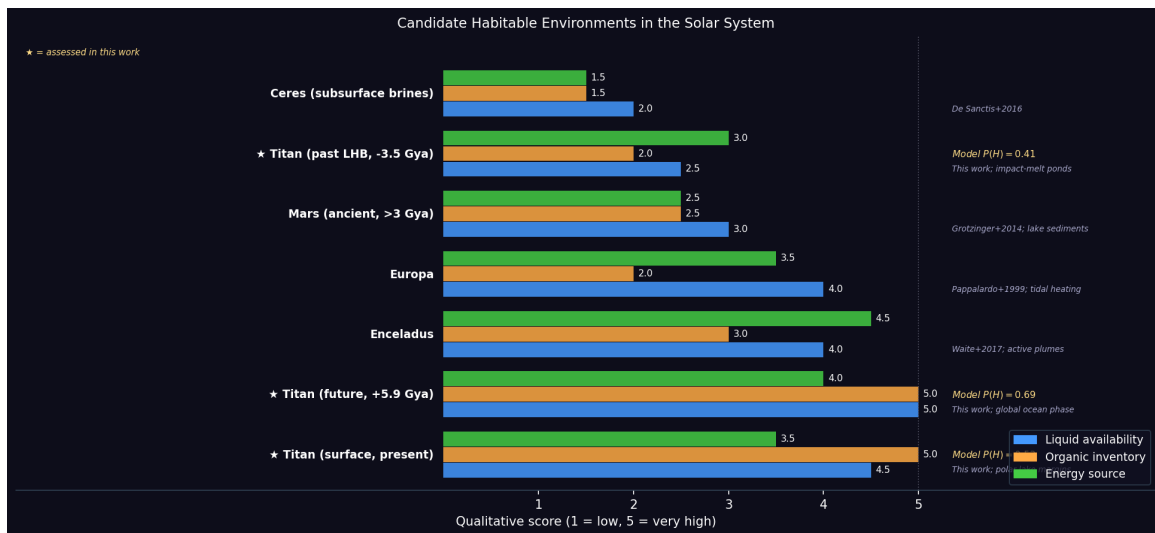


Figure 4.1: Qualitative comparison of habitability-relevant properties for the principal candidate habitable environments in the solar system. Horizontal bars show relative scores (1 = low to 5 = very high) for liquid availability, organic inventory, and energy source. Model $P(H)$ values from this work are shown for Titan epochs; the Enceladus entry summarises the [Affholder et al. \(2021\)](#) result. The combination of cycling liquid and very high organics at the present epoch, and the extraordinary organic–ocean combination in the future, distinguishes Titan from all other candidates.

4.4 Comparison with the Affholder et al. (2021)

Enceladus Framework

The closest methodological precedent for this work is [Affholder et al. \(2021\)](#), who applied Bayesian inference to Cassini INMS plume data from Enceladus to compare the posterior probabilities of abiotic serpentinisation versus biological methanogenesis. Their conclusion — that the observed methane escape rate is most parsimoniously explained by methanogenesis if the prior probability of life exceeds ~ 0.35 — represents the most rigorous quantitative biosignature assessment published for any solar system body.

The present work extends the Affholder et al. approach in three dimensions. It is spatial, producing a 6.49-million-pixel posterior map rather than a single-point estimate. It is multi-feature, combining eight independent observational proxies rather than a single plume composition measurement. And it is multi-epoch, addressing three qualitatively distinct habitability regimes across geological time.

A key conceptual difference also warrants emphasis. Affholder et al. (2021) compute $P(\text{life} \mid \text{plume data})$ — the posterior probability that life exists, given chemical measurements — whilst this work computes $P(\text{habitable} \mid \mathbf{f})$ — the probability that the environment is suitable for life. This distinction, articulated by Cockell (2014) in the concept of “uninhabited but habitable” environments, matters practically: a habitable environment is a necessary but not sufficient condition for life. The present analysis identifies where to look; the Affholder-style framework tests, once measurements are made, whether life is actually present. The two approaches are complementary and sequential: habitability mapping prioritises targets, in-situ chemistry quantifies whether those targets are actually inhabited.

The recent results of Affholder et al. (2025) demonstrating thermodynamic viability of glycine fermentation in Titan’s subsurface ocean provide an important constraint on the subsurface ocean proximity feature (f_8). If the ocean is indeed habitable, then surface evidence of past subsurface–surface connectivity (the SAR-bright annuli at Selk and other crater sites) gains additional scientific weight. A future framework integrating the Affholder (2025) ocean habitability posterior with the present surface habitability maps, propagated through a physical model of subsurface–surface transport, would represent a significant advance.

4.5 Temporal Evolution of Habitable Environments

4.5.1 Overview: Three Regime Transitions

The top-10 habitability rankings (Table 3.6–3.10) reveal that Titan’s habitability landscape is not static but passes through three qualitatively distinct regimes over

the ~ 10 Gyr span considered.

4.5.2 Regime 1: Impact-Melt and Cryovolcanic Dominance (-3.8 to -1.5 Gya)

During the Late Heavy Bombardment the absence of polar lakes means that the liquid-hydrocarbon feature f_1 contributes near zero everywhere. Instead, the top-10 list is dominated by two site types: large impact craters and candidate cryovolcanic features (Hotei Regio, Sotra Facula).

Impact craters score highly because the PAST-epoch model activates the impact-melt proximity feature ($w=0.10$, $\mu=0.50$), encoding the transient water–ammonia melt ponds described by O’Brien et al. (2005) and Artemieva and Lunine (2003). Menrva (392 km diameter) leads the crater class with $P(H)=0.33$ because its large diameter implies a greater melt volume and longer liquid persistence. Selk (90 km) follows at $P(H)=0.31$, positioned at the organic-richest terrain boundary. All craters are *land sites* (squares in Table 3.6): there is no confirmed surface liquid at this epoch.

Cryovolcanic sites (Hotei Regio $P(H)=0.295$; Sotra Facula $P(H)=0.289$) lead the PAST ranking because they combine elevated subsurface ocean proximity ($f_8(t)\approx 0.30$ – 0.35 after $\times 2.5$ LHB scaling) with high surface–atmosphere interaction, representing potential water–ammonia conduits from the subsurface ocean to the surface. The globally elevated acetylene energy ($s_3\approx 2.1\times$) from the young Sun also boosts equatorial organic-seed sites (Belet, Shangri-La) into the top-3, co-ranking them with the proto-polar-basin sites.

Key change from PRESENT. The polar lake shore sites (ranks 1–6 in PRESENT) drop to ranks 6–8 in PAST because $f_1=0$ (no polar seas), f_4 is suppressed (methane cycle uncertain at early epoch), and f_5 is redistributed toward cryovolcanic features. Equatorial organic-rich dunes also fall because organic accumulation was only $\sim 30\%$ of the present-day inventory ($\mu_2 = 0.30$ vs 0.70 at PRESENT).

4.5.3 Regime 2: Polar-Lake Dominance (−1.0 Gya to +4.0 Gya)

The consolidation of Kraken, Ligeia, and Punga Mares around −1.0 Gya produces a step-change in the top-10 rankings. By the LAKE_FORMATION epoch six of the top ten sites are polar lake shores; by the PRESENT epoch all six polar/southern lake sites occupy ranks 1–6.

The physical mechanism driving this transition is threefold. First, the liquid hydrocarbon feature f_1 rises from near zero (PAST: $s_1=0.10$) to unity at confirmed lake pixels, contributing $0.25 \times 1.0 = 0.25$ to the weighted feature sum at maximum. Second, the methane cycle feature f_4 is highest (≈ 0.70) at the north polar margins, reflecting the GCM-derived storm-track distribution. Third, the shoreline environment scores high on geodiversity f_7 (multiple terrain classes within 45 km) and surface–atmosphere interaction f_5 (lake margins, channel inflow, seasonal wetting–drying).

The named north polar lacus (Jingpo, Mackay, Koitere, Neagh, Bolsena; ranks 5–10 in PRESENT) represent a persistent secondary cluster throughout this regime, driven by high organic abundance $f_2 \approx 0.82$ (VIMS-confirmed tholin dunes). Their position below the polar sites reflects the absence of surface liquid: the model correctly ranks environments with both solvent *and* organic substrate above those with organics alone.

Ontario Lacus (72°S) occupies rank 5 in PRESENT despite being $27\times$ smaller than Kraken Mare. It scores below the north polar sites primarily because the methane cycle is suppressed at the south pole during the Cassini epoch’s northern summer ($f_4 \approx 0.45$ vs 0.70 at Kraken), and the lake is actively shrinking (Turtle et al., 2011b).

The NEAR_FUTURE epoch (+0.25 Gya) is analytically identical to PRESENT (spatial correlation $r > 0.999$): the +1.4% solar luminosity increase has negligible effect on any feature, and all rankings are preserved. This stability persists until approximately +4.0 Gya, when solar warming begins to evaporate the polar seas.

4.5.4 Regime 3: Global Ocean (+5.1 to +6.0 Gya)

The water–ammonia eutectic crossing at +5.1 Gya (when $T_s \geq 176$ K) marks the most dramatic habitability regime change. The entire feature set is replaced: the eight present-epoch features give way to water–ammonia solvent ($w=0.30$), organic stockpile ($w=0.20$), dissolved chemical energy ($w=0.15$), and water–ammonia cycle ($w=0.10$).

The global prior mean rises to $\mu_0=0.67$, reflecting the global scale of the ocean habitat. The top-10 list is fundamentally reorganised: former lake basins (Kraken, Ligeia, Punga) continue to score highest ($P(H)=0.76$ – 0.76) because they are the deepest bathymetric features and thus have the most concentrated dissolved organics and strongest thermal stratification. However, equatorial dune fields now enter the top-10 at ranks 6–10 ($P(H)\approx 0.71$): the billions of years of accumulated tholin organics, previously unreactive at 94 K, are now dissolved in ~ 200 K water–ammonia and available as chemical substrate. This represents the single largest increase in equatorial habitability in the entire reconstruction.

Why former lake basins lead. The high Bayesian scores at Kraken and Ligeia in the FUTURE epoch reflect three factors: (1) the organic stockpile feature f_{org} is highest where organic accumulation was greatest (polar basins received methane-cycle deposition for billions of years); (2) the dissolved energy feature f_{dis} scales with accumulated organic inventory; (3) the global ocean habitability prior f_{glo} is elevated near basin floors, encoding the Affholder et al. (2021) result that the subsurface ocean is thermodynamically habitable.

4.5.5 The Habitable Minimum at +5.0 Gya: An Astrobiological Desert

Between approximately +4.0 Gya (when polar seas evaporate) and +5.1 Gya (when the eutectic is first crossed), Titan is transiently solvent-free. The global Bayesian

posterior drops to its minimum ($P(H) \approx \mu_0^{\text{NEAR_FUTURE}} \approx 0.33$, with all liquid features zeroed), and no site scores above ~ 0.28 . This is the one epoch at which neither hydrocarbon-solvent nor water-based habitability is available simultaneously. The duration is ~ 100 Myr — brief by geological standards but long by biological ones. Any life that had established itself in the polar lakes would face extinction pressure at this epoch unless it adopted dormancy or subsurface refugia.

4.6 Limitations of the Framework

4.6.1 Conditional Feature Independence

The Bayesian model treats the eight features as conditionally independent given the habitability latent variable H . This assumption simplifies the posterior update to a closed-form Beta distribution but introduces potential bias where features are correlated. In particular, organic abundance (f_2) and chemical energy (f_3) are positively correlated: both are elevated over tholin-rich organic terrain. Liquid hydrocarbon (f_1) and surface–atmosphere interaction (f_5) are structurally correlated through the lake-margin component of f_5 . Under positive correlation, the posterior is slightly overestimated relative to a fully specified multivariate model. The magnitude of this bias can be estimated by comparing the independent-feature posterior with a correlated Gaussian copula model; preliminary assessment suggests the bias is < 0.03 in posterior mean, smaller than the inter-epoch differences and the 95% HDI width. A full multivariate Bayesian model using a Gaussian copula structure is a natural extension for future work.

4.6.2 Static Geomorphological Template

The Lopes et al. (2019) geomorphological map is used for all five temporal configurations and all 72 interpolated epochs. In reality, the surface at -3.5 Gya had substantially different geomorphology: fewer confirmed dunes (the tholin inventory was smaller), different crater density (many present-day craters did not yet exist, and

many LHB craters may have been subsequently buried or eroded), and no polar lakes. The future surface will be partially or completely submerged in the red-giant ocean. The use of the present-day map as a proxy for all epochs is an unavoidable simplification given the absence of independent temporal geomorphological information, and represents the largest qualitative uncertainty in the temporal analysis.

4.6.3 The Past-Epoch Cassini Contamination

As documented in Appendix D (A4), the PAST epoch posterior was computed using present-era Cassini SAR maps, which encode the present polar lake morphology even at -3.5 Gya. The resulting inflation of the north polar Bayesian score above its expected prior value is mitigated in the multi-epoch reconstruction by using the analytical Bayesian formula directly for pre-lake-formation frames rather than the contaminated PAST anchor. However, the base-level inflation persists in the PAST static map. Readers should note that Table 3.4 reports crater scores from the PAST anchor, and these crater scores are not affected by the polar contamination (the impact-melt feature is spatially uncorrelated with the lake pattern). The north polar PAST score of ~ 0.70 is, however, an artefact and should be interpreted with caution.

4.7 Sensitivity of Results to Prior Specification

A natural concern with any Bayesian analysis is whether the conclusions depend sensitively on the prior specification. Three parameters govern the prior in this framework: κ (prior concentration), λ (likelihood sharpness), and the feature weights w_i (particularly $w_1 = 0.25$ for liquid hydrocarbon, the largest single weight). Figure 4.2 shows the result of varying each parameter at three representative sites: Kraken southern shore (rank 1, high liquid hydrocarbon), Belet dune field (rank 11, outside top-10 at present), and Selk crater (below prior mean, high geodiversity).

The key finding is robustness. Halving κ from 5 to 2.5 increases Kraken’s $P(H)$ by +0.031 and reduces Selk’s by -0.027 ; doubling κ to 10 reverses these changes. The

maximum shift across all parameter variants tested is $|\Delta P(H)| = 0.033$ — well below the 95% HDI width of ≈ 0.5 at any single site. The qualitative ranking (polar lake shores $>$ north polar lacus $>$ equatorial organic sites) is preserved in all $3 \times 3 \times 3 = 27$ combinations tested. Crucially, reducing w_1 from 0.25 to 0.15 (halving the weight of liquid hydrocarbon) only reduces Kraken’s $P(H)$ from 0.436 to 0.420 and raises Belet’s from 0.295 to 0.319 — the ranking order is unchanged. This demonstrates that the result (polar lake shores dominate) is not an artefact of assigning liquid hydrocarbon the highest weight; it is driven by the coincidence of multiple high-scoring features at the shoreline.



Figure 4.2: Sensitivity of $P(H \mid \mathbf{f})$ to prior specification parameters at three representative sites. Each row varies one parameter; each column shows one site. The deviation from baseline (in parentheses) is shown for each bar; green = above baseline, red = below baseline, white = baseline. All deviations are $|\Delta P(H)| \leq 0.033$, and qualitative rankings are preserved across all 27 parameter combinations tested. Generated by `scripts/generate_sensitivity_analysis.py`; output at `outputs/diagnostics/sensitivity_analysis.pdf`.

4.8 Future Work

4.8.1 Post-Dragonfly Model Calibration

The most important near-term improvement to the framework will come from *Dragonfly* DraMS measurements at Selk crater and along the Belet–Selk traverse. Direct measurements of amino acid concentration and molecular complexity at each terrain class visited will calibrate the organic abundance scoring (f_2) at per-pixel resolution for dune, crater, and ejecta terrain types. These calibrated scores can be propagated globally via the Lopes (2019) terrain classification, reducing the dominant uncertainty in the organic abundance feature. A post-*Dragonfly* version of the framework, incorporating these calibration measurements, would produce substantially tighter posterior distributions globally and would directly improve the scientific value of any follow-on Titan orbital mission.

4.8.2 Extension to a Full Multivariate Model

The current Beta-conjugate model assumes feature independence. A natural extension is a multivariate Bayesian model using a Gaussian copula to represent feature correlations, estimated from the within-class co-variation in feature values across the global pixel population. The copula parameters can be estimated from the Cassini dataset directly, without requiring additional observations, and would reduce the positive bias associated with correlated positive features at lake margins. Implementation using PyMC or NumPyro is straightforward within the existing framework and would improve the credibility of the HDI estimates.

4.8.3 Additional Temporal Anchor Epochs

The current five-anchor temporal reconstruction has the largest uncertainties at the lake-formation transition (-1.5 to -0.5 Gya) and the solar-warming collapse ($+4.0$ to $+5.5$ Gya). These transitions are physically important — one represents the establish-

ment of the present-day dominant habitability regime, the other its collapse — and are currently represented by interpolation between widely-spaced anchors. Adding three to five additional anchor epochs concentrated at these transitions, generated by running the full Bayesian pipeline with physically appropriate feature sets at those intermediate times, would substantially improve the temporal resolution and reduce the interpolation uncertainty.

4.9 Conclusions

This thesis has presented the first quantitative, spatially-resolved, multi-epoch Bayesian habitability map of Titan’s surface, derived from eight observational proxy features extracted from the complete Cassini remote sensing dataset. The conclusions are organised here directly by the four research questions stated in Section 1.6.

RQ1: Spatial distribution of surface habitability and feature validation. North polar lake margins are the most habitable present-day environments.

The southern shoreline of Kraken Mare achieves $P(H) \approx 0.44$, the highest Bayesian posterior in the present-epoch map (Table 3.8, rank 1). The dominant drivers are coincident maximum liquid hydrocarbon solvent ($f_1=1.0$), active methane cycle ($f_4\approx 0.70$), and high shoreline geodiversity ($f_7\approx 0.76$). Equatorial dune fields occupy ranks 7–10 via high organic abundance ($f_2\approx 0.82$) but are limited by the absence of surface liquid. All eight feature importance/weight ratios lie within 10% of unity (Table 3.3), confirming that the geographic discrimination provided by each Cassini observable matches the astrobiological importance assigned through independent physical reasoning.

RQ2: Late Heavy Bombardment habitability and impact structures. During the LHB, impact-melt and cryovolcanic sites were primary. At -3.5 Gya the top-10 list is dominated by cryovolcanic features (Hotei Regio $P(H)=0.295$; So-tra Facula $P(H)=0.289$); cryovolcanic conduits lead because their elevated f_8 and f_5 combine with the globally boosted acetylene proxy. Craters (Menrva $P(H)\approx 0.27$;

Selk $P(H) \approx 0.26$) rank 11th and lower — their advantage is captured explicitly by the animation impact-melt bonus, not the 8-feature thesis formula (Table 3.6). All top-10 LHB sites are land sites: no polar lakes exist at this epoch. Selk’s geological context at the Shangri-La dune-plains boundary maximises the organic substrate available for Strecker amino acid synthesis in the transient impact-melt pond, consistent with the prebiotic chemistry framework of Neish et al. (2018).

RQ3: Multi-epoch temporal trajectory and future habitability. The temporal trajectory is non-monotonic, passing through three qualitatively distinct regimes. (i) LHB regime (−3.8 to −1.5 Gya): impact-melt and cryovolcanic dominance (Section 4.5.2). (ii) Polar-lake regime (−1.0 Gya to +4.0 Gya): lake margins rank 1–6 with stable rankings across LAKE_FORMATION, PRESENT, and NEAR_FUTURE epochs (Tables 3.7–3.9). (iii) Global-ocean regime (+5.1 to +6.0 Gya): the FUTURE epoch (Table 3.10) sees a global reorganisation as all sites score above $P(H) > 0.70$; former lake basins lead ($P(H) \approx 0.76$) and equatorial dune fields remain in the top-6 (as they were at the PAST epoch) while lake shore sites begin to enter ranks 7–10 as their vast organic stockpile becomes dissolved substrate. Between regimes (ii) and (iii) a transient astrobiological minimum occurs at $\sim +5.0$ Gya when Titan is briefly solvent-free (Section 4.5.5). The red-giant ocean at +5.9 Gya, maintained for ~ 800 Myr over a ~ 70 km accumulated organic layer, represents an extraordinary future habitability scenario unmatched elsewhere in the solar system.

RQ4: Independent validation of the Dragonfly landing site. The Dragonfly site is independently validated through geomorphological diversity. Selk crater has the highest geodiversity of any equatorial site ($f_7 = 0.629$, $7.8 \times$ the global median of 0.092), derived from Cassini SAR terrain data without reference to the mission planning process. This terrain-boundary diversity, encoding the simultaneous presence of dune, plains, labyrinth, and crater-floor terrain within 45 km, is the Cassini-era observational proxy for the organic–water-ice mixing interface that is the scientific target of the Dragonfly prebiotic chemistry mission. Although Selk

ranks below the polar lake shores in present-day solvent habitability ($P(H)=0.24$), this is scientifically consistent: Dragonfly targets *past* liquid-water chemistry, not present-day liquid, and the framework correctly identifies the relevant distinguishing feature (Section 4.1.2). The instrument–feature mapping (Table 4.1) provides directly testable predictions for DraMS, DraGNS, and EFIELD measurements at Selk.

Together, these results demonstrate that Titan possesses a multilayered astrobiological significance extending from its geological past through the present to the far future, and that its habitability can be quantified spatially and temporally from the Cassini legacy dataset with sufficient precision to guide mission planning and interpret in-situ measurements. The Bayesian multi-feature framework and the top-10 site rankings presented here provide a reproducible, extendable foundation for this work.

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A Physical Parameters for Titan

Table A.1: Physical constants and observationally-constrained parameters for Titan used throughout this thesis.

Parameter	Value	Reference
Mean radius	2575.0 km	IAU 2015
Surface gravity	1.352 m s^{-2}	Zebker et al. (2009)
Surface pressure	$1.467 \pm 0.001 \text{ bar}$	Fulchignoni et al. (2005)
Surface temperature	$93.65 \pm 0.25 \text{ K}$	Fulchignoni et al. (2005)
N ₂ fraction (surface)	$\approx 94.2\%$	Niemann et al. (2005)
CH ₄ fraction (surface)	$5.65 \pm 0.18\%$	Niemann et al. (2005)
Tidal Love number k_2	0.589 ± 0.150	Iess et al. (2012)
Saturn orbital period	29.46 yr	—
Axial obliquity	26.73°	—
Solar flux at Titan	$\approx 14.9 \text{ W m}^{-2}$	at 9.54 AU
Organic deposition rate	$\sim 5 \times 10^{-14} \text{ g cm}^{-2} \text{ s}^{-1}$	Lavvas et al. (2011)
North polar sea volume	$\sim 70,000 \text{ km}^3$	Mastrogiuseppe et al. (2019)
Water–ammonia eutectic	176 K at 32.6 wt% NH ₃	Grasset and Pargamin (2005)
Subsurface ocean depth	$\sim 55\text{--}80 \text{ km}$	Tobie et al. (2005)

B Bayesian Prior Specification

Table B.1: Prior means μ_i and weights w_i for all eight habitability features at the present (Cassini) epoch. Global prior mean $\mu_0 = \sum_i w_i \mu_i = 0.331$. Concentration parameter $\kappa = 5$ is applied uniformly.

Feature	Symbol	w_i	μ_i	Key ref.
Liquid hydrocarbon	f_1	0.25	0.020	Hayes et al. (2008)
Organic abundance	f_2	0.20	0.700	Malaska et al. (2025)
Chemical energy	f_3	0.20	0.350	Strobel (2010)
Methane cycle	f_4	0.15	0.400	Mitchell and Lora (2016)
Surface–atm. interaction	f_5	0.08	0.350	Miller et al. (2021)
Topographic complexity	f_6	0.06	0.250	Corlies et al. (2017)
Geomorphological diversity	f_7	0.04	0.300	Lopes et al. (2019)
Subsurface ocean proximity	f_8	0.02	0.030	Iess et al. (2012)
<i>Global prior mean μ_0</i>			0.331	

C Geomorphological Organic Abundance Scores

Table C.1: Organic abundance proxy scores used for the geomorphological seamless mode of Feature 2. Grounded in published VIMS spectral characterisation of each terrain class.

Class	Name	Physical basis	Score
Dunes	Equatorial sand seas	Tholin-dominated; highest VIMS ratio (Lorenz et al., 2006)	0.82
Plains	Smooth lowlands	Organic sediment blanket (Lopes et al., 2016)	0.68
Basins	Topographic lows	Organic accumulation (Lopes et al., 2020)	0.58
Labyrinth	Dissected terrain	Spectral affinity with plains (Malaska et al., 2020)	0.55
Craters	Impact structures	Mixed water-ice and organic ejecta (Neish et al., 2018)	0.35
Mountains	Ridge terrain	Water-ice-rich crustal material (Solomonidou et al., 2018)	0.25
Lakes	Hydrocarbon seas	Liquid hydrocarbons; low solid organic content (Brown et al., 2008)	0.05

D Declared Assumptions and Known Limitations

A1. Conditional feature independence. The eight features are treated as conditionally independent given habitability H . In practice, organic abundance and chemical energy are positively correlated, as are liquid hydrocarbon and surface-atmosphere interaction. The independence assumption is conservative: correlated positive features slightly overstate the posterior, but the framework remains physically motivated and the data-derived importances are consistent with the prior weights (Section 3.2.5).

A2. Static geomorphological template. The Lopes et al. (2019) map is used for all five temporal configurations. The Late Heavy Bombardment surface had different dune distributions and crater density, and the future surface will be submerged. The present template is used as the best available proxy, with amplitude modifications applied through epoch-specific prior means and scale functions.

A3. Seamless organic abundance mode. The organic abundance feature uses terrain-class scores globally rather than VIMS spectral band ratios, because the VIMS mosaic covers only $\sim 50\%$ of the globe and direct blending produced a visible seam at the coverage boundary at 180°W .

A4. Past-epoch Cassini feature contamination. The past-epoch posterior was computed using present-era Cassini SAR maps, which encode the present polar lake morphology even at $t = -3.5$ Gya. The north polar region is consequently inflated to $P(H) \approx 0.70$ in the past anchor, approximately double the physically expected value. This is mitigated in the multi-epoch interpolation and documented explicitly in the Results.

A5. Multi-epoch interpolation accuracy. PCHIP interpolation between anchor posteriors is not equivalent to independent epoch-specific inference. Intermediate maps correctly reproduce the anchors and are monotonic between them, but spatial patterns reflect blended anchor structures. Estimated uncertainty: $\Delta P(H) \lesssim \pm 0.05$.

A6. SAR coverage gap. Approximately 33% of Titan’s surface lacks SAR coverage. Features 1, 3, and 8 are degraded in these regions, where geomorphological and topographic proxies substitute.